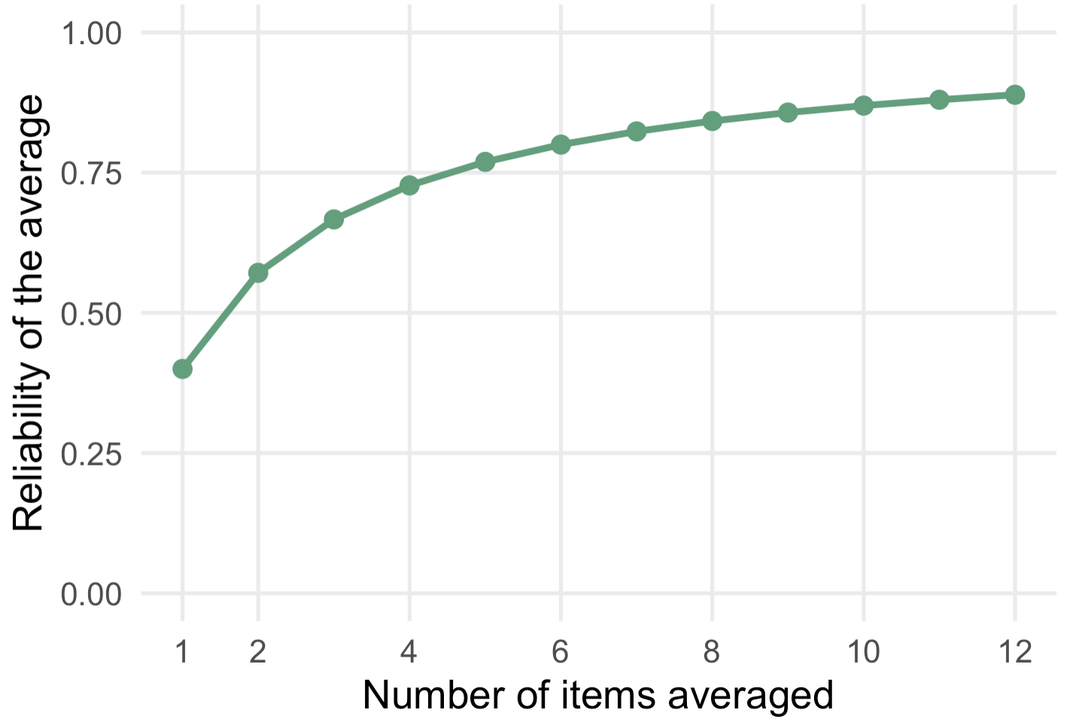
Measurement II

Latent variables

Sebastian Ramirez-Ruiz
European University Institute (EUI)
November 6, 2026

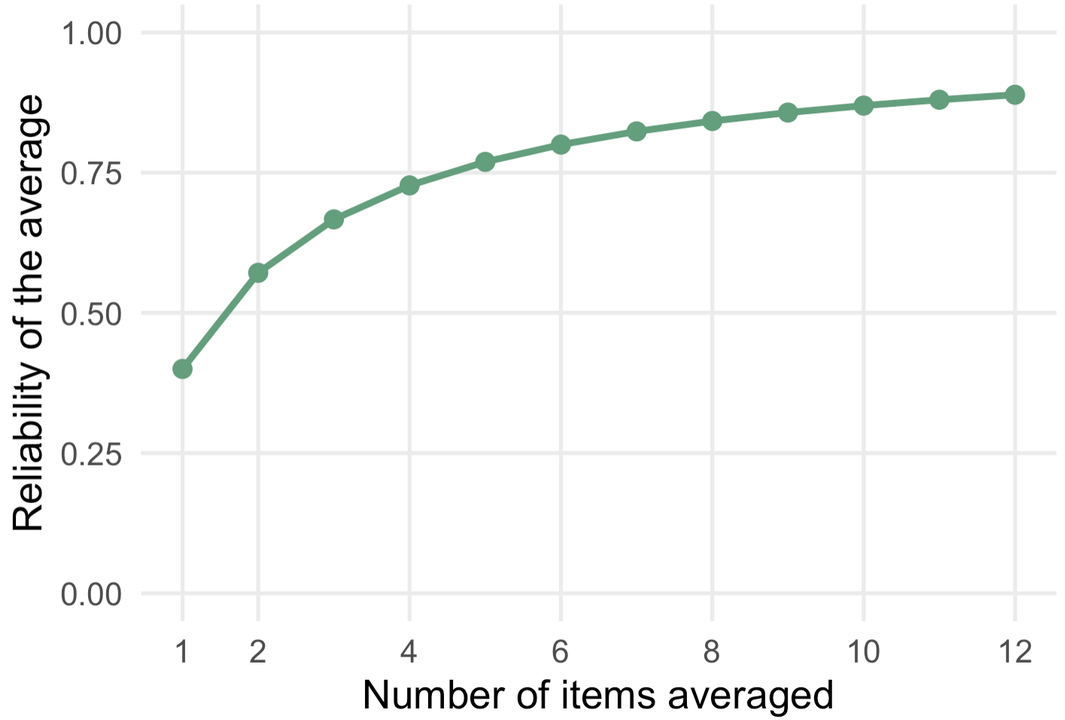
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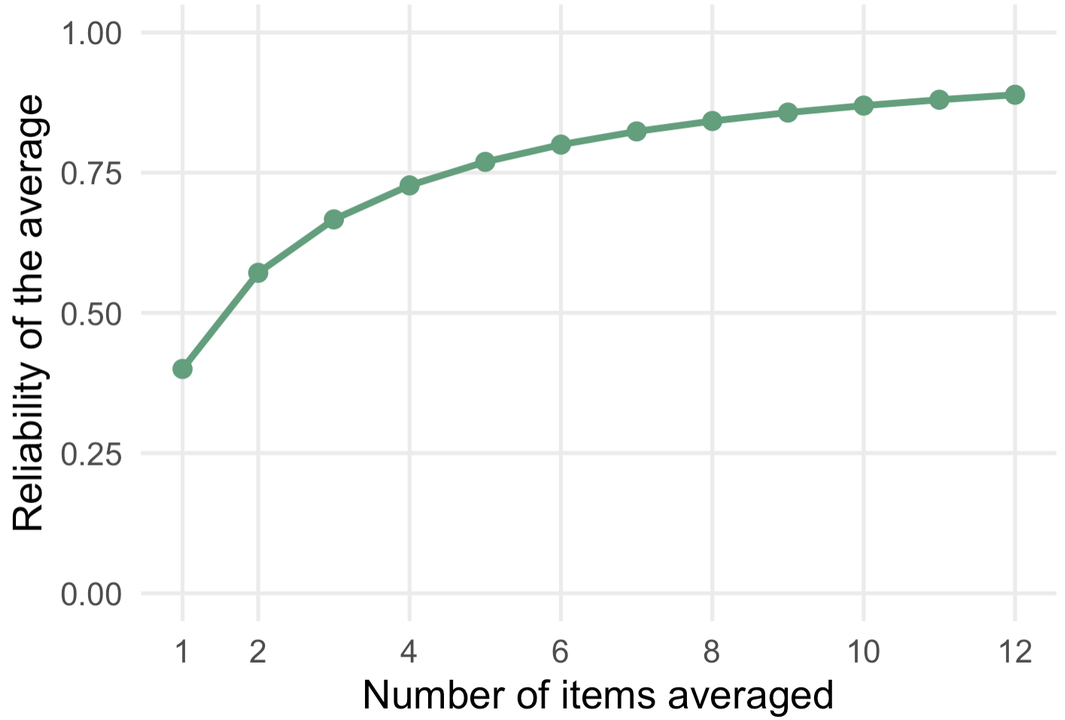
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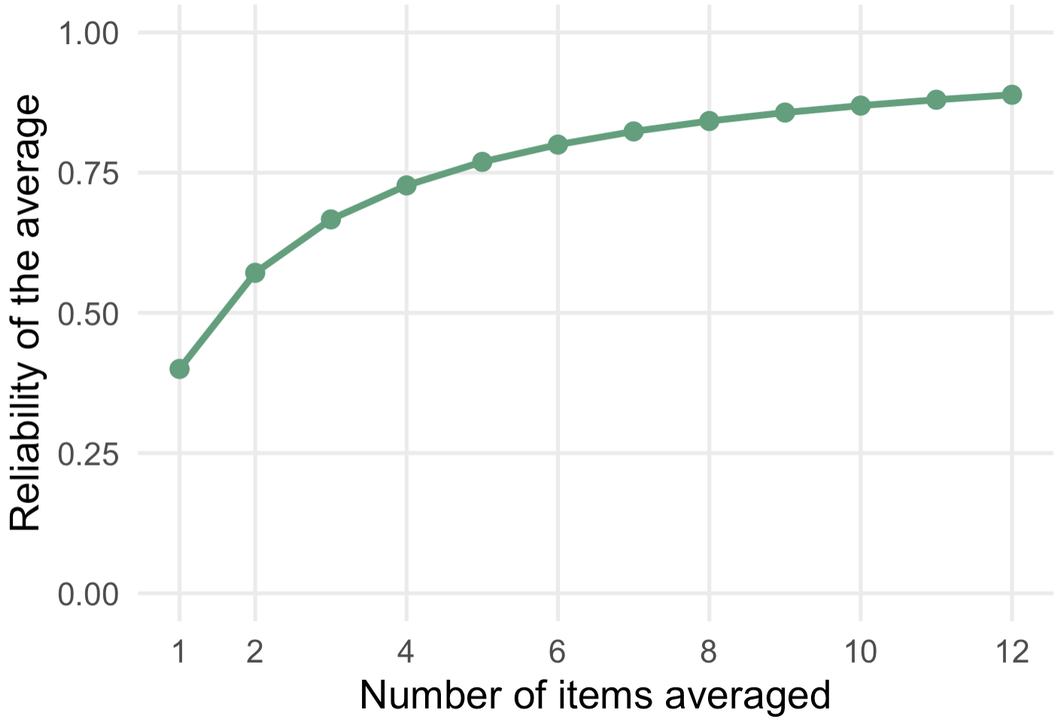
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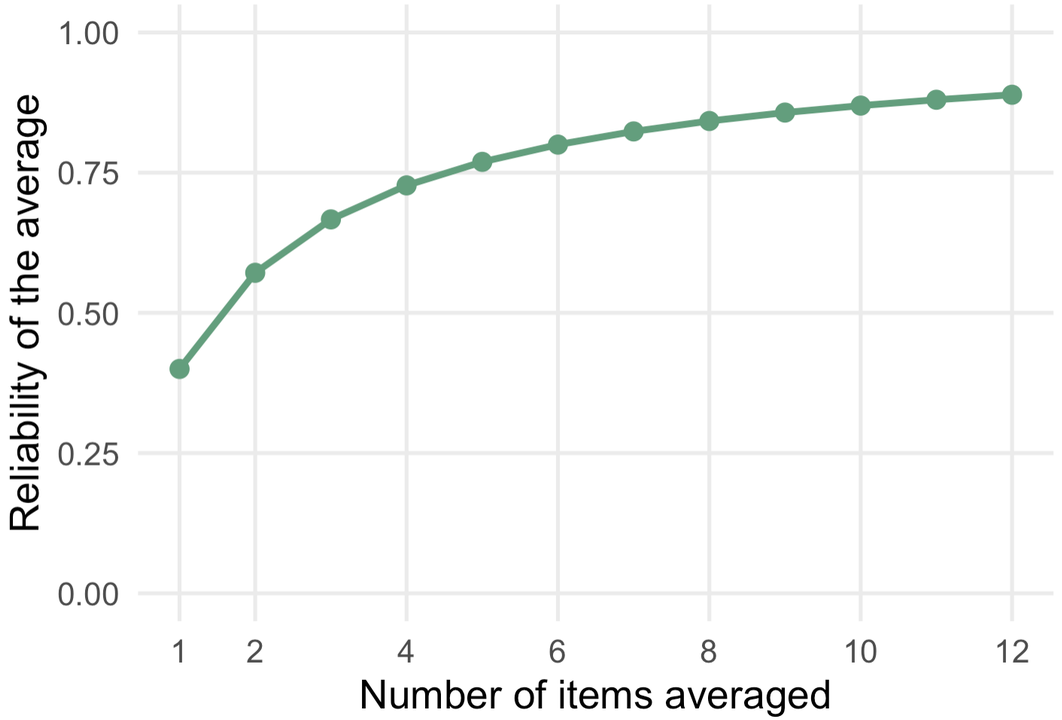
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- In Lab 4 you built a **trust in institutions** scale from six items and computed Cronbach's α by hand. The police item looked weak, and the lab warned: "α is a lower bound".
- Today: replace "average and hope" with **a model** of how a construct produces its items, and find out what the police item was telling us.



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Where today fits

Week 4 listed five sources of validity evidence. Today is mostly about one of them.

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- A latent-variable model is a **theory of internal structure** that the data can contradict. That is what makes it evidence, and not just a way to average.

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Practice

Two exercises today, ending with the bones of a measurement paper for your PhD; factor models on the trust battery in Lab 5

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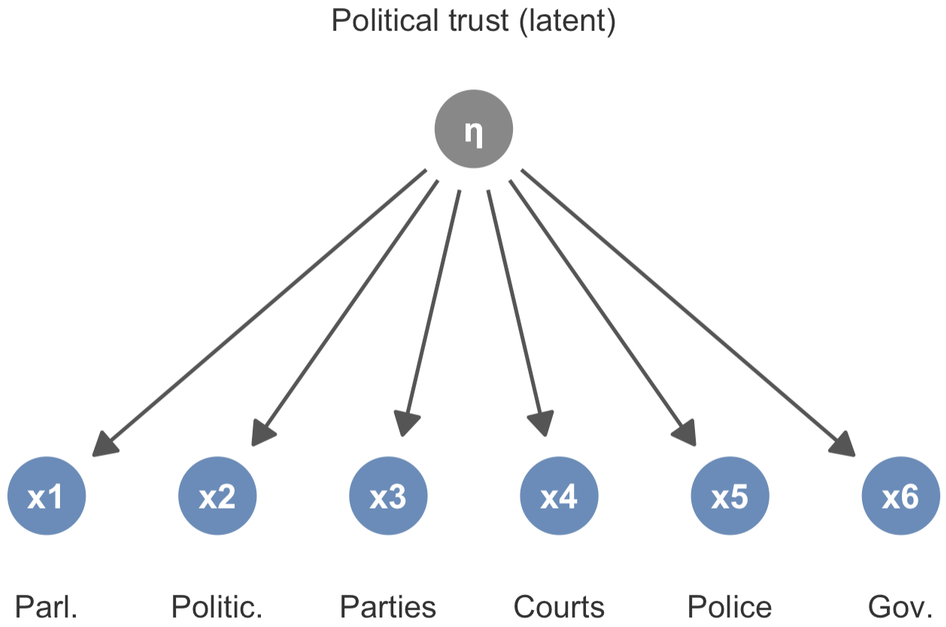
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- Before any model: **which trust** is your theory about?

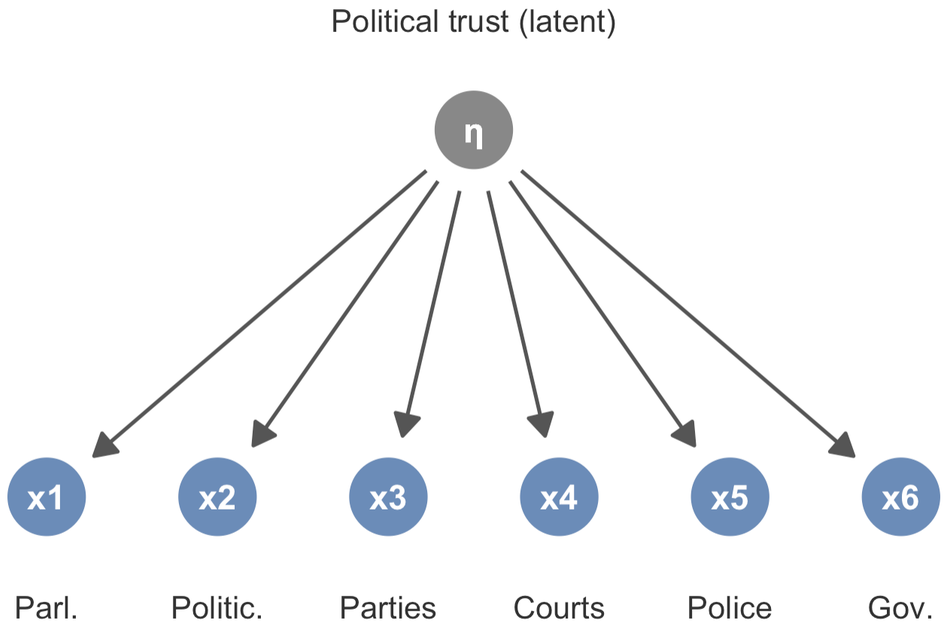
Political trust as a latent variable



One latent attitude (gray, unobserved) causes six survey answers (blue, measured). Unique errors are left out of the drawing.

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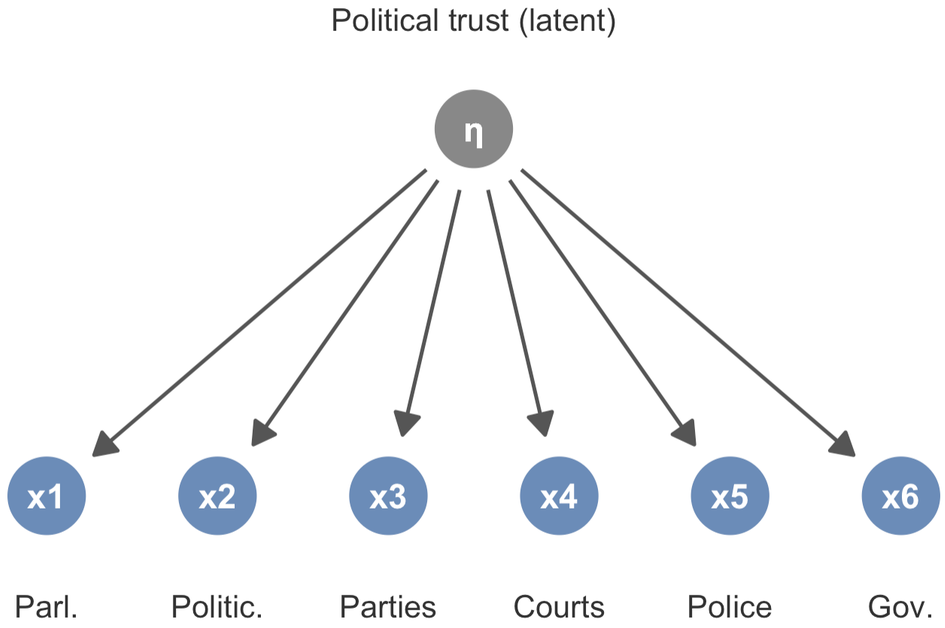
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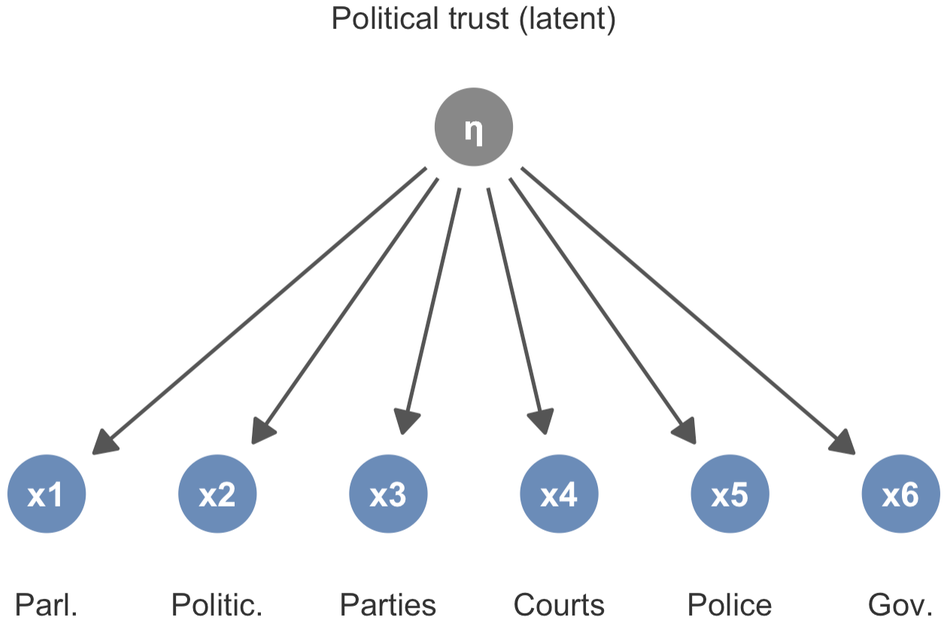
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- A latent-variable model is a **DAG with an unobserved cause**: the same picture as Week 4's measurement-error DAGs, with many arrows out of one node.



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The measurement model

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- We never observe η_i, but we can estimate the λ_j from how the items covary. Week 4's $D^* = D + e$ is the special case with one item and $\lambda = 1$.

Why many items beat one



Correlations among eight simulated trust items (0–10). All positive, but courts, police, and the civil service form their own block.

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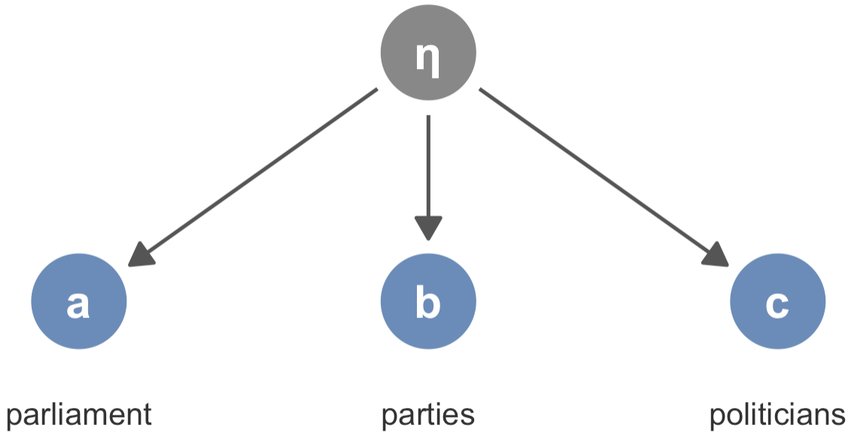
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- Look at the matrix: courts, police, and the civil service correlate **more with each other** than one "trust" can explain. Local independence fails for a one-factor model: a hint of a second factor.



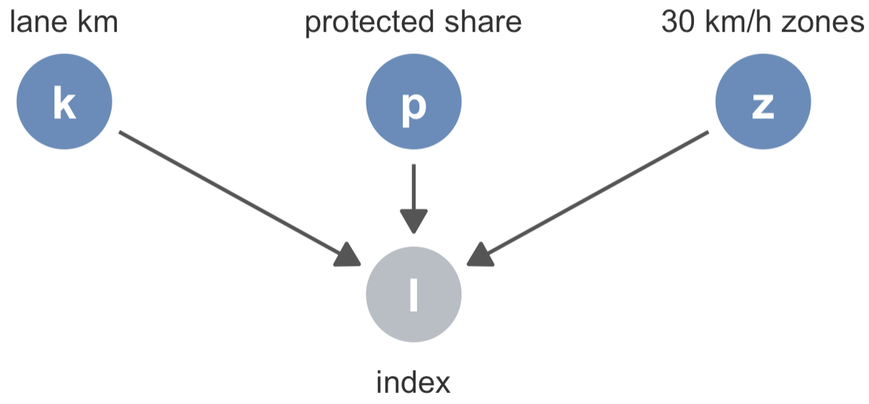
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Reflective vs formative

Reflective: political trust

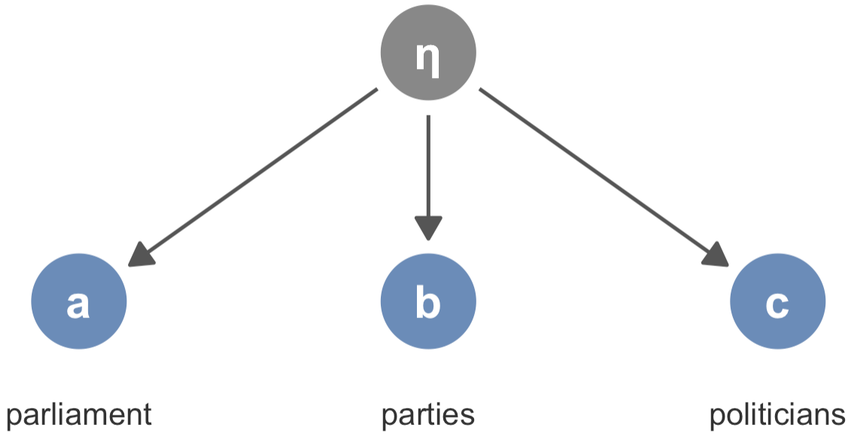


Formative: a 'bikeability' index

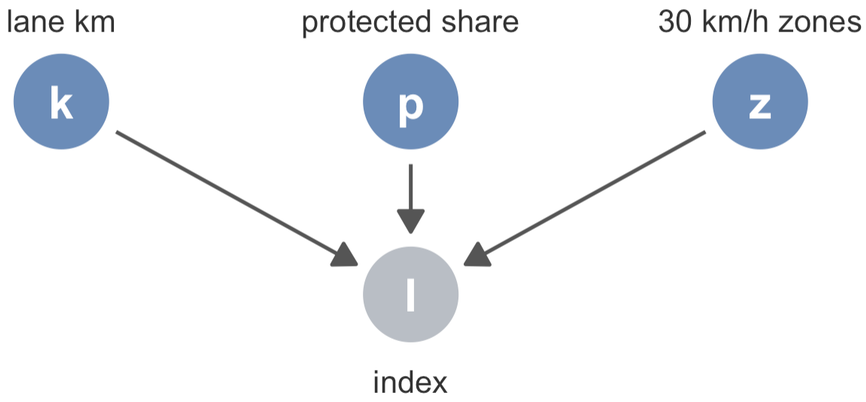


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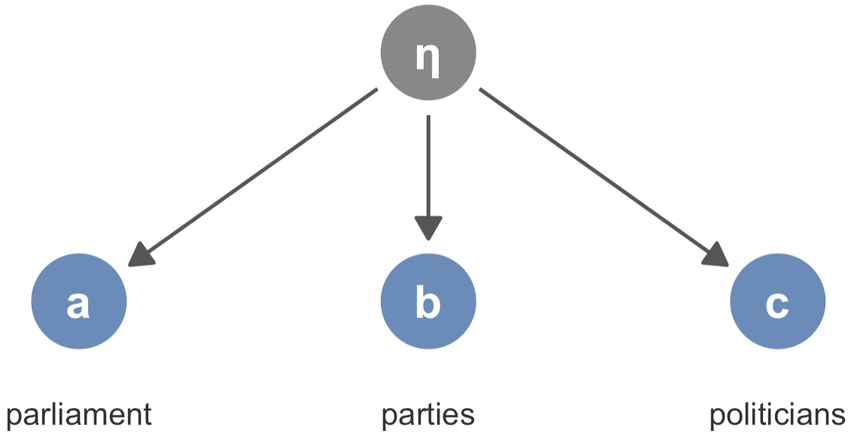
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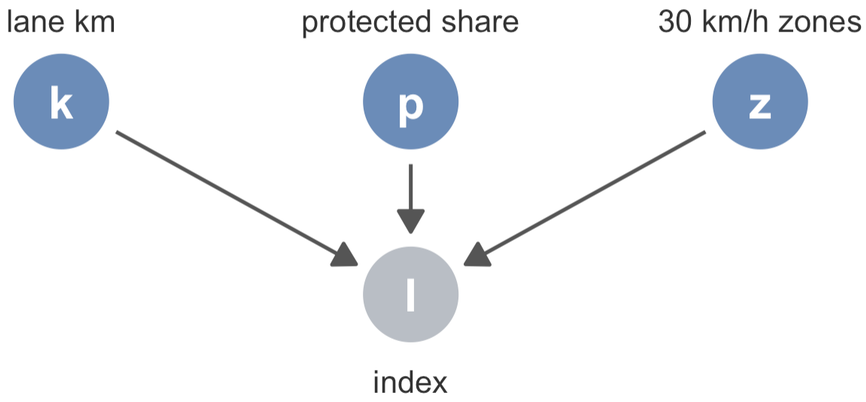
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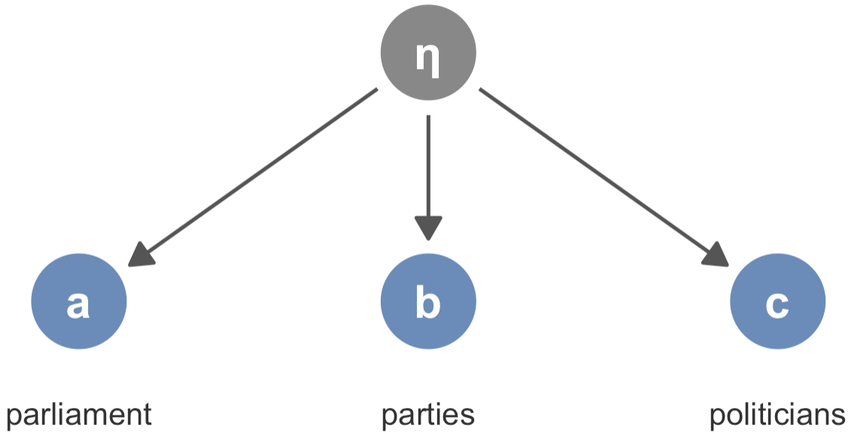
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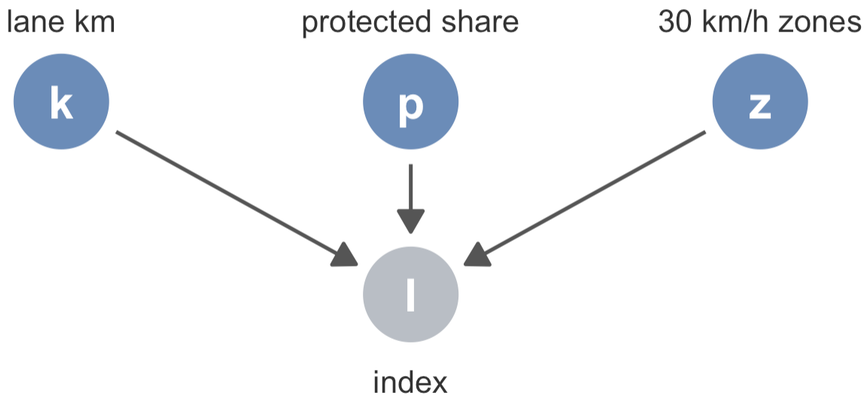
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- Factor models assume the reflective story. **For a formative index, they are the wrong theory.**

PCA is not factor analysis

Compress the data, or model the construct?

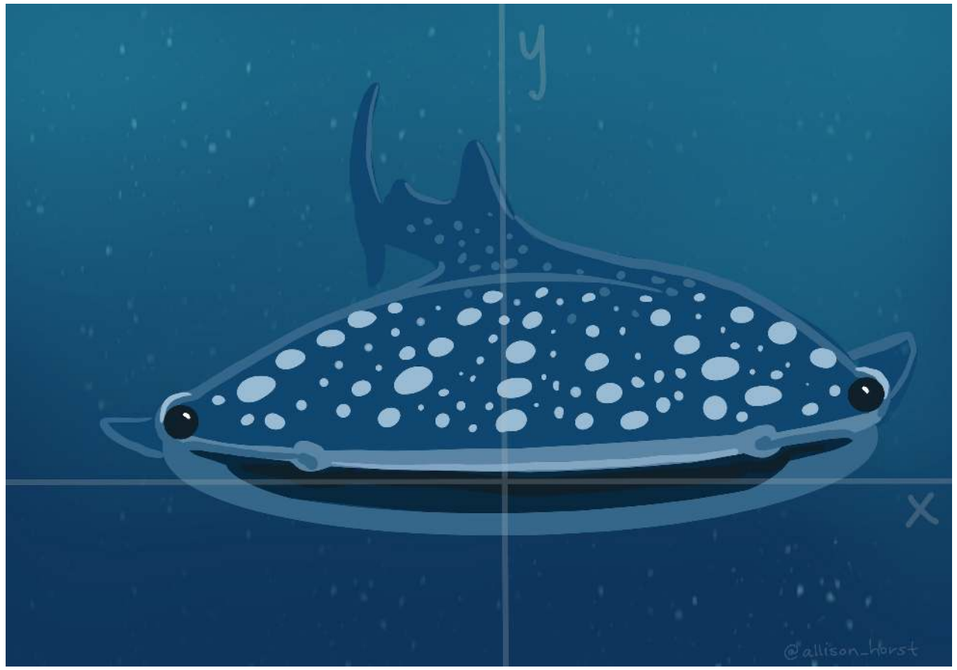
A swarm of krill spreads along a diagonal (correlated data). PCA turns the axes so the first one runs along the widest spread, the way the wide-mouthed shark lines up to catch the most. Artwork by Allison Horst (@allison_horst).

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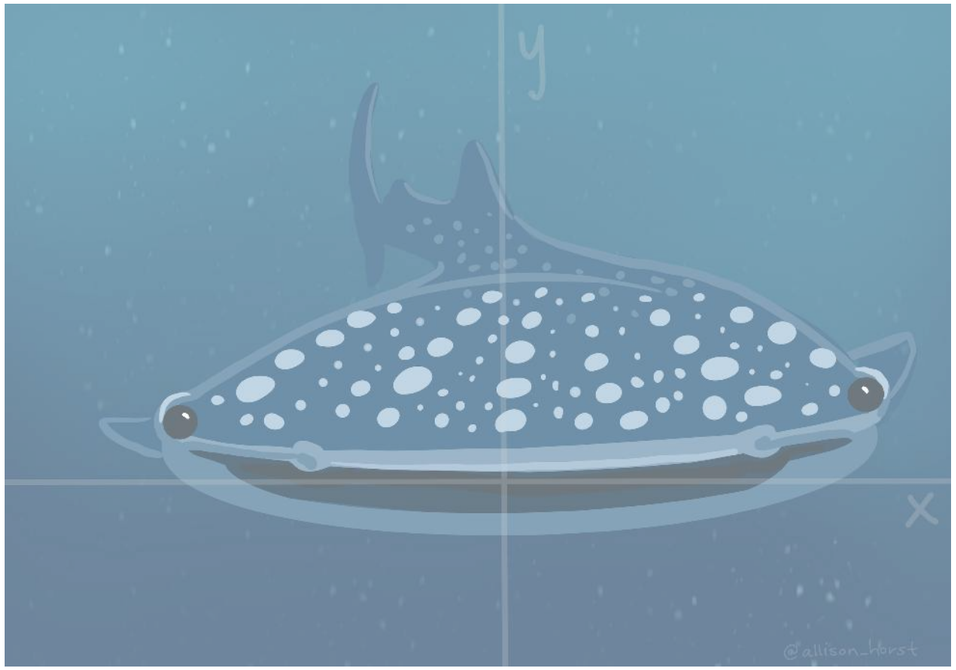
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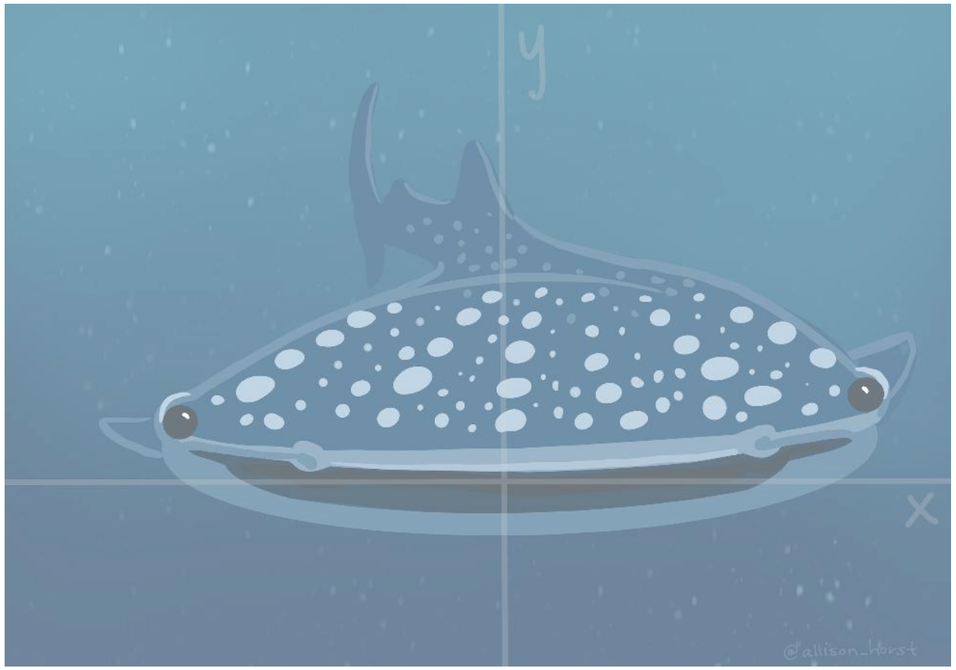
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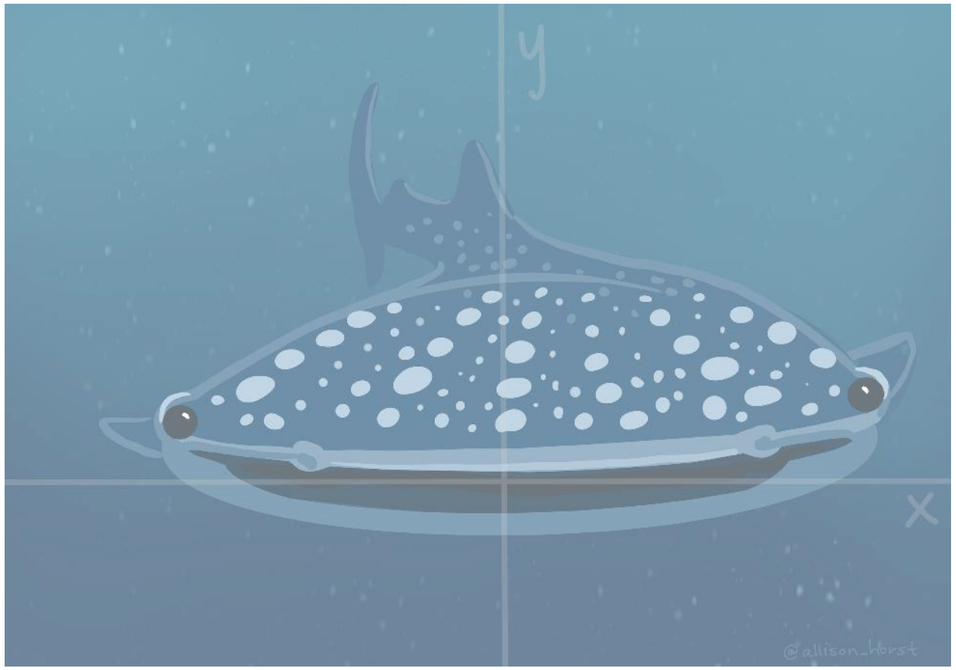
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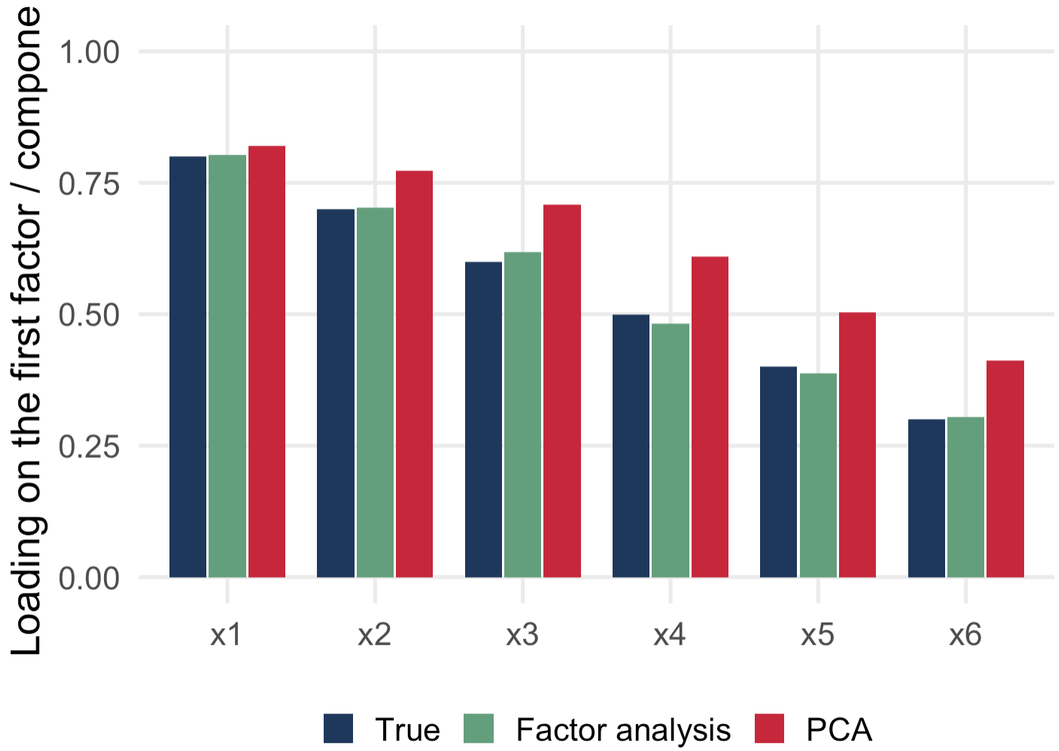
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- Factor analysis has a **theory of error**; PCA does not. One is a measurement model, the other a summary of the data for **dimension reduction**.



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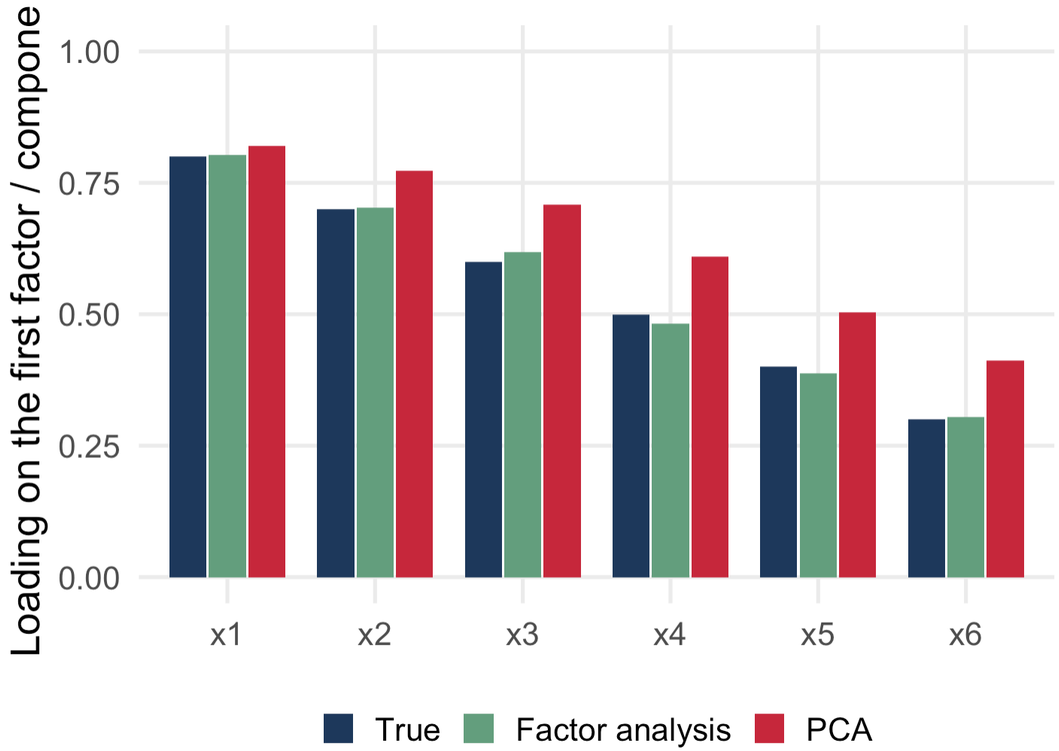
Same data, different answers



Six simulated items whose true loadings (navy) fall from .8 to .3. Factor analysis (green) recovers them; PCA (red) inflates the weak ones.

Same data, different answers

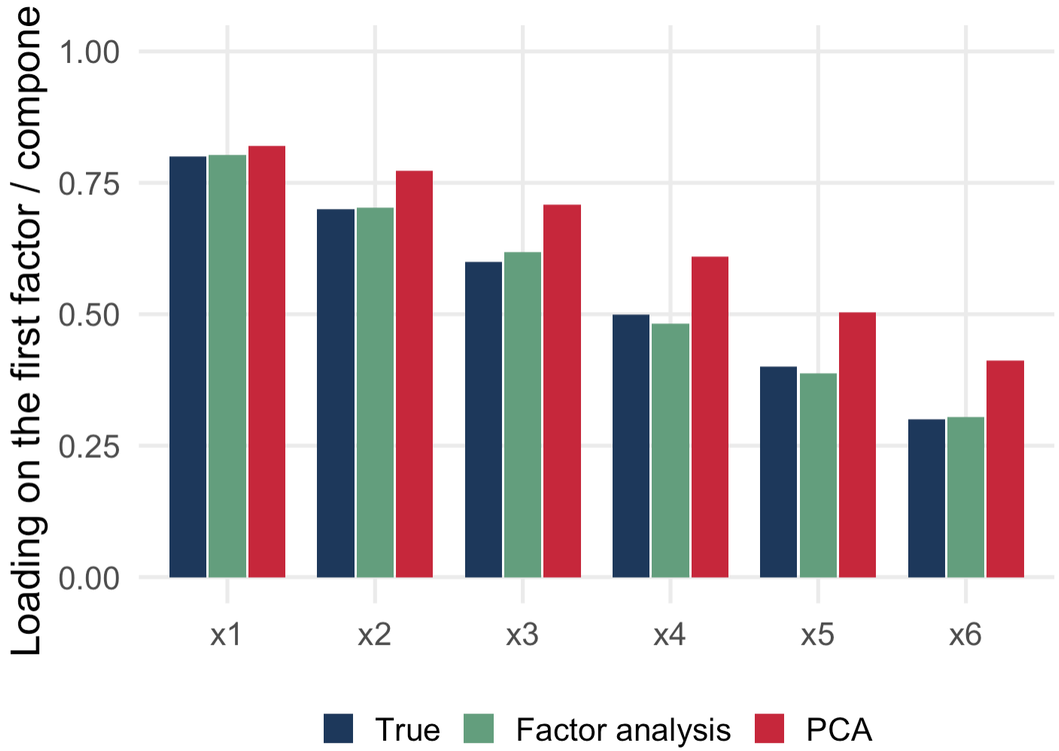
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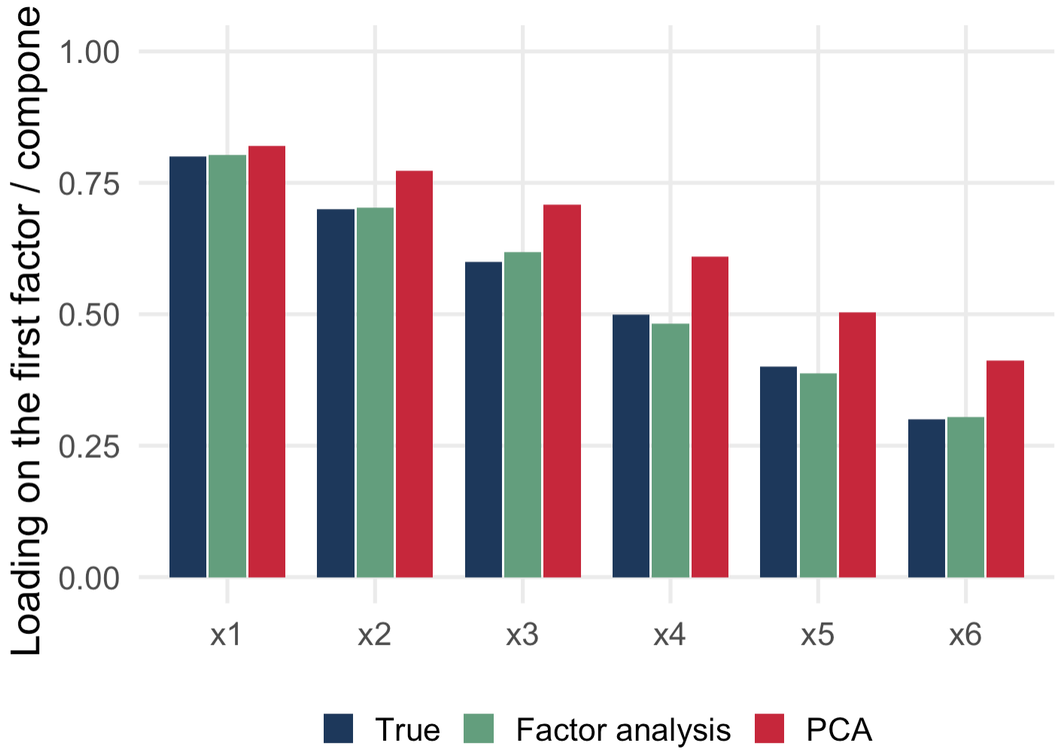
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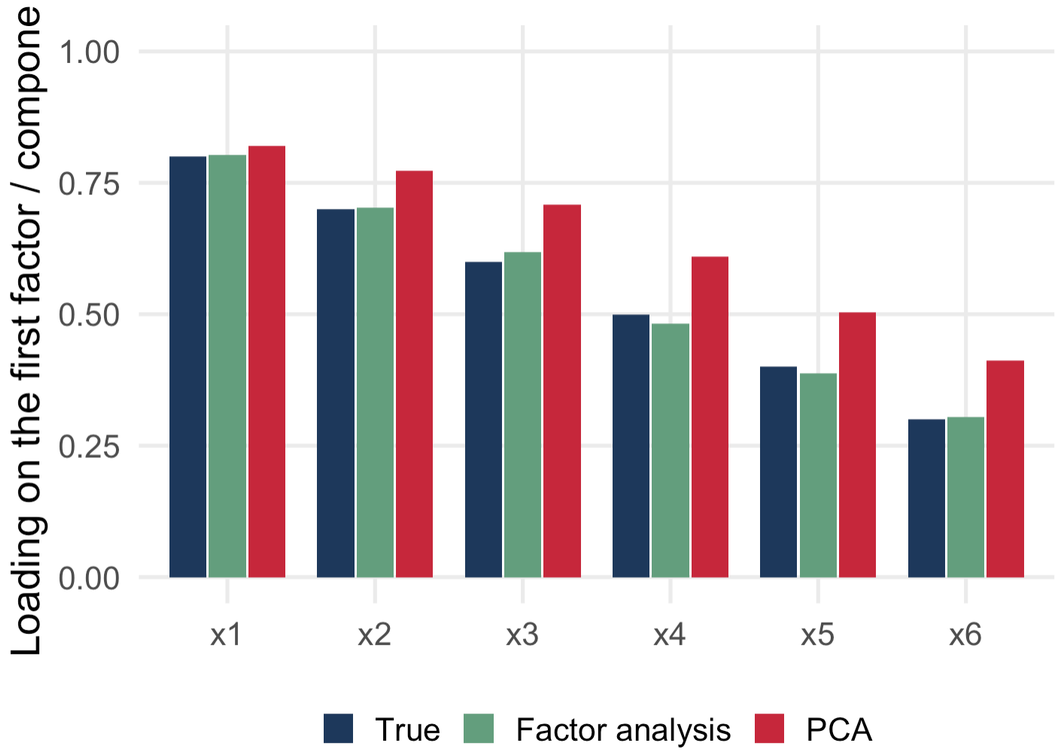
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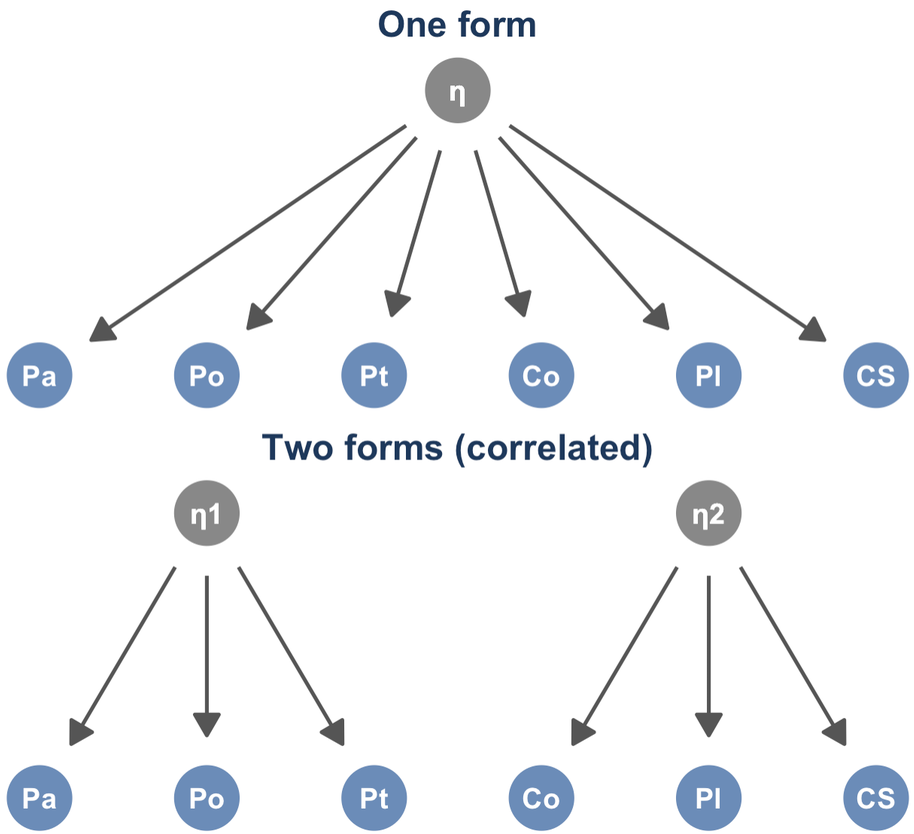
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- With few items or weak loadings, the difference is large; with many strong items, it shrinks.



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One form of political trust, or two?

A debate that a model can settle*

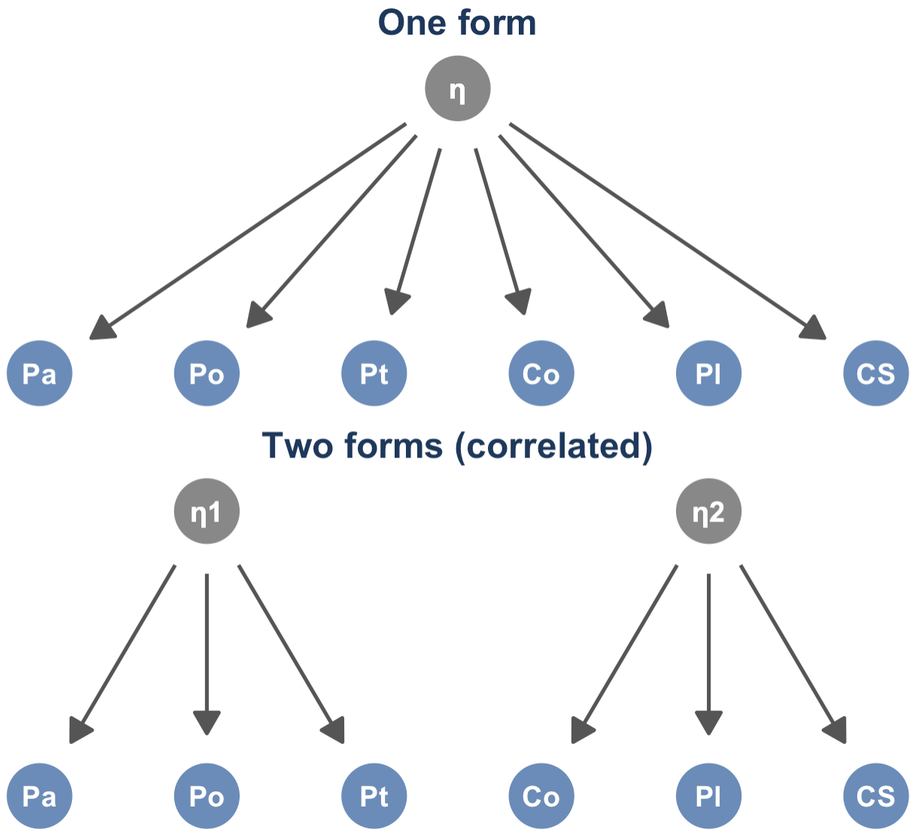


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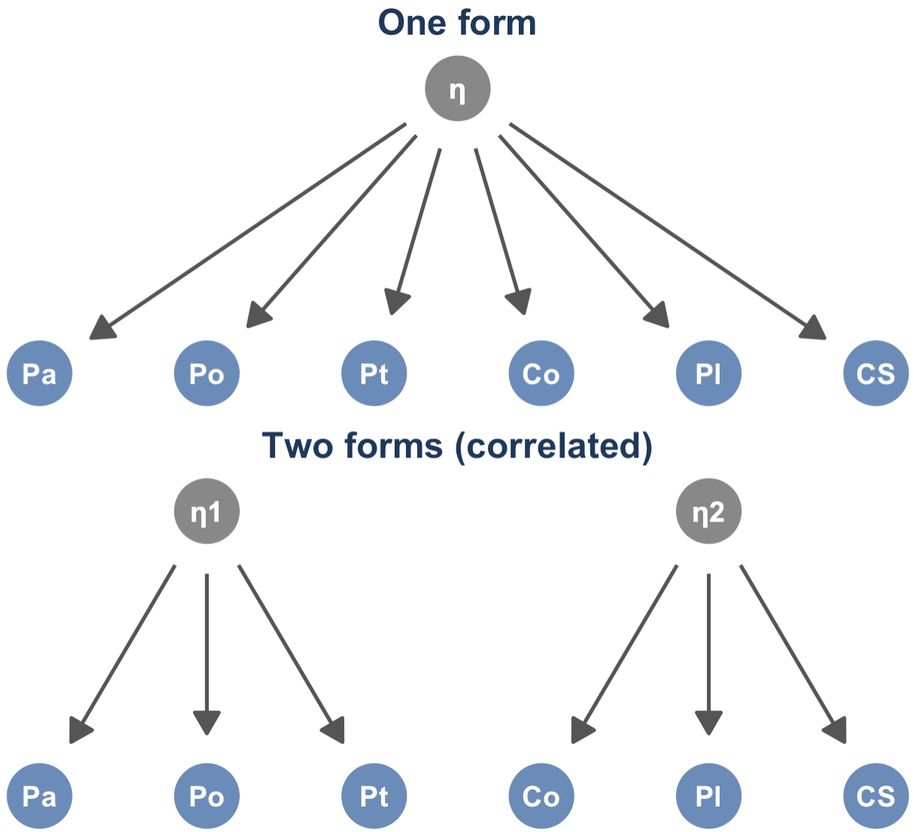


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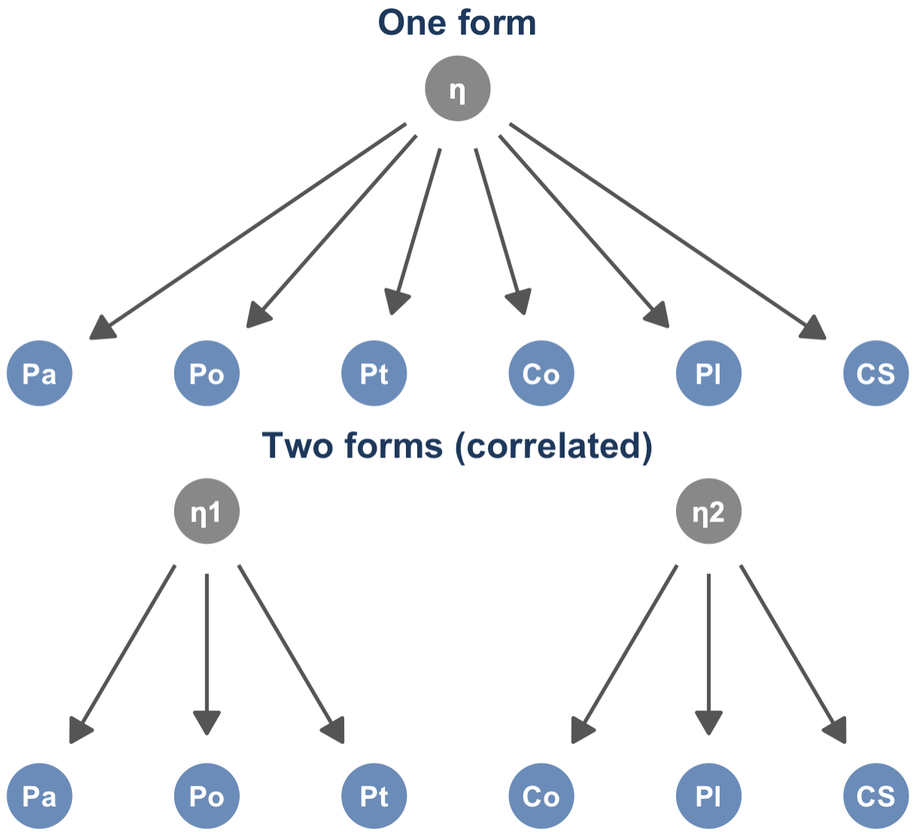


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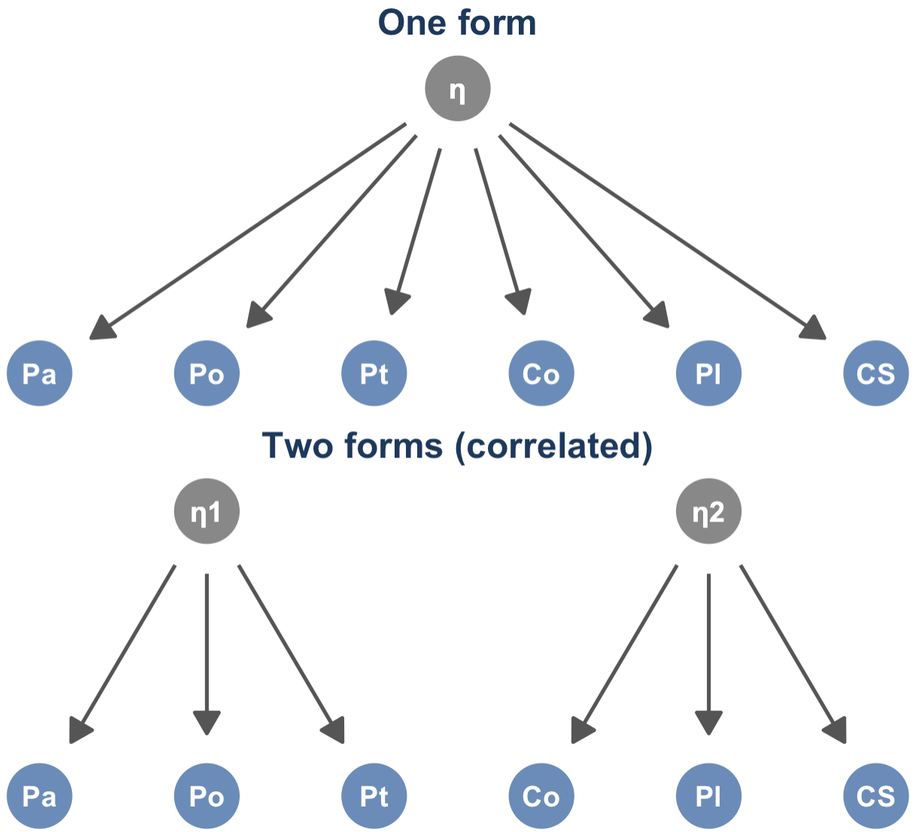


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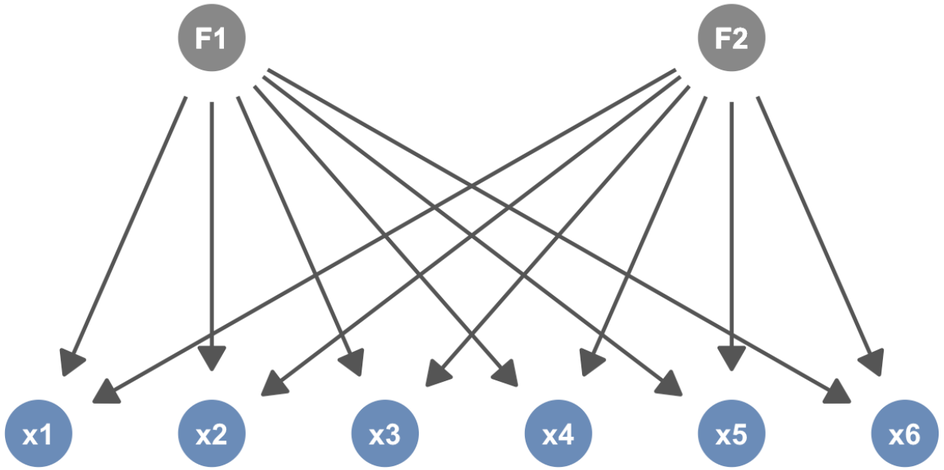


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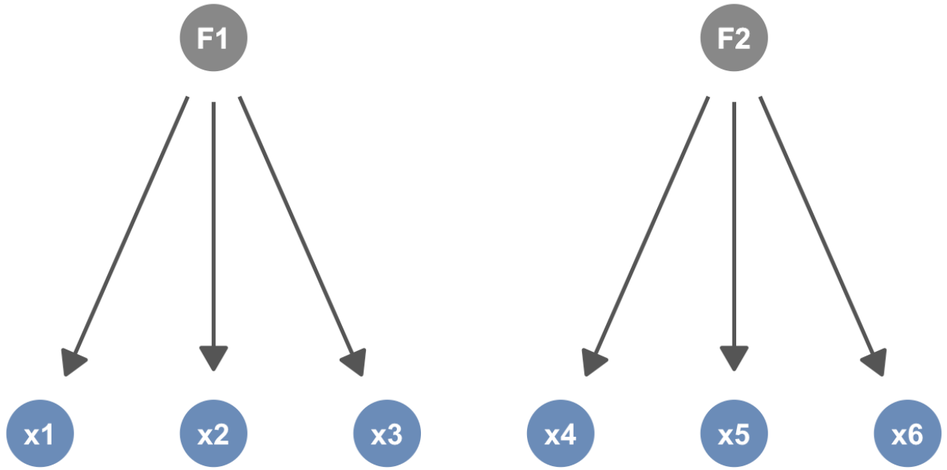
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EFA or CFA? What and when

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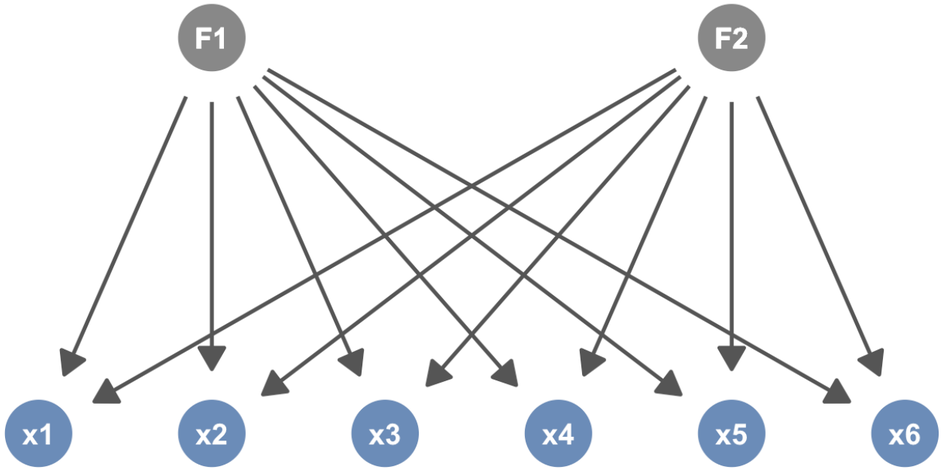


Confirmatory: you decide which arrows exist

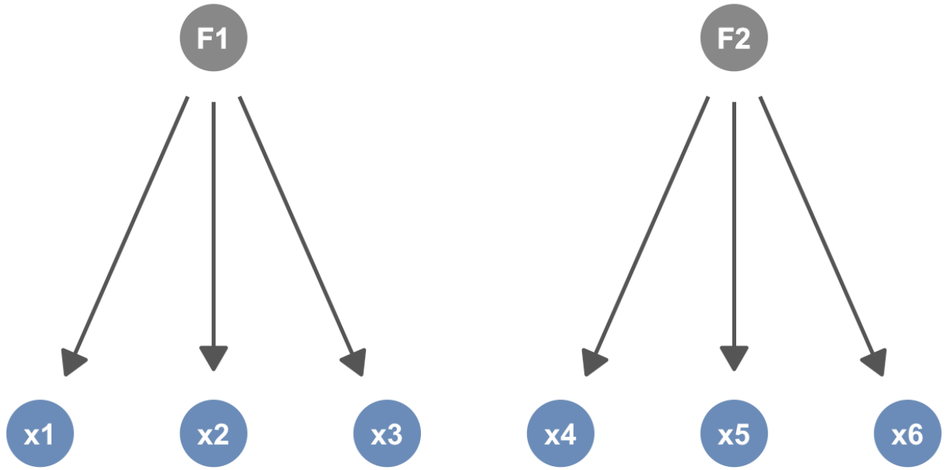


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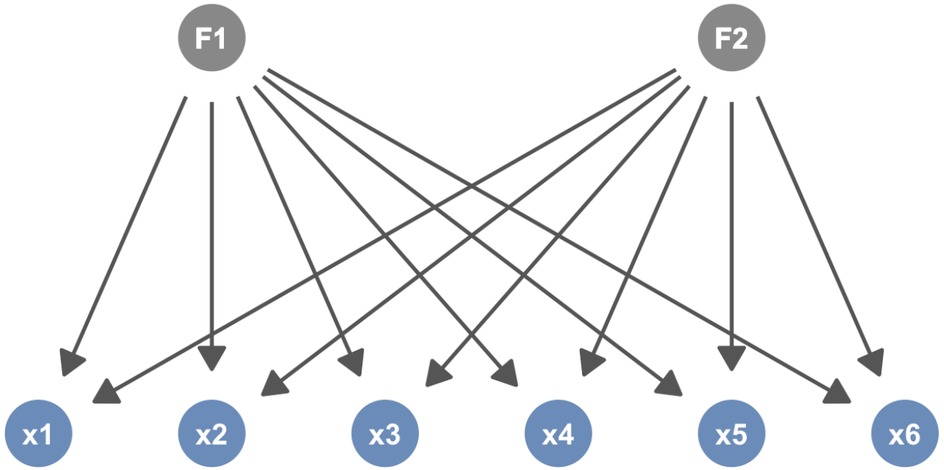


EFA: let the data suggest a structure

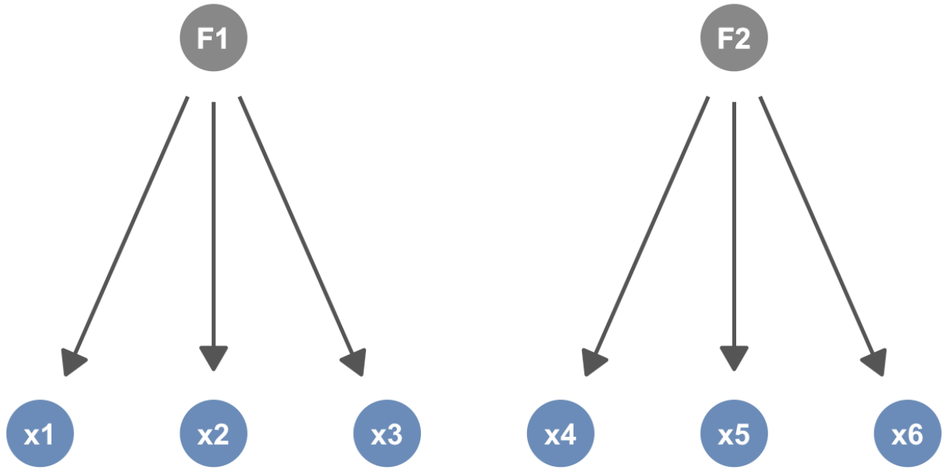
Use it **early**: a new pool of items, a weak theory about how many dimensions there are. Example: Campos & Federico's 45 draft items.

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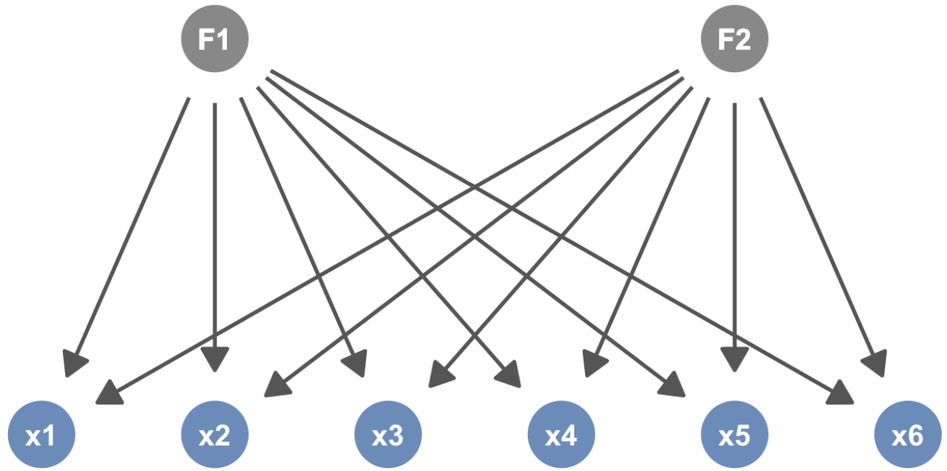
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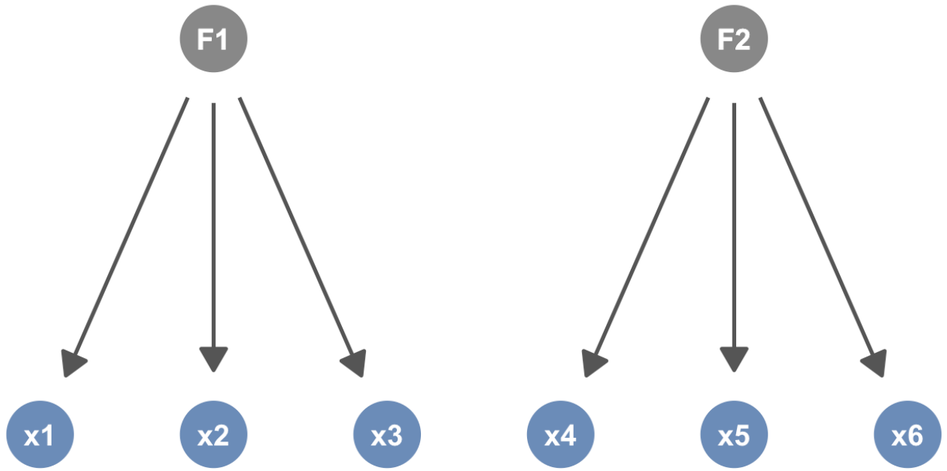
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Exploring and confirming on the **same** data is not confirmation.

Exploratory factor analysis in four steps

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Extract

Estimate the common factors. Maximum likelihood assumes roughly normal items and, in return, gives a test of fit. Principal-axis factoring assumes nothing about the distribution, but gives no fit test. With Likert items, results are usually similar.

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Eigenvalues above 1, the "elbow" of a scree plot, parallel analysis. Look for several criteria that agree, and factors you can interpret.

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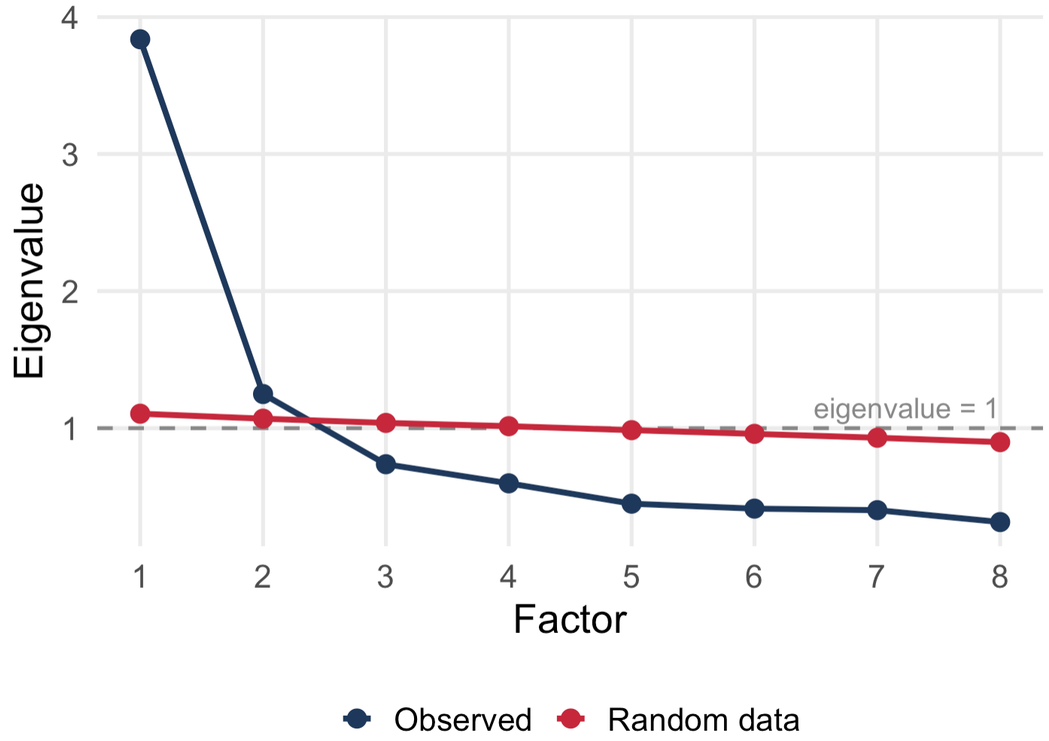
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- Every step is a decision. Steps 2 and 4 are where most of the **forking paths** live.

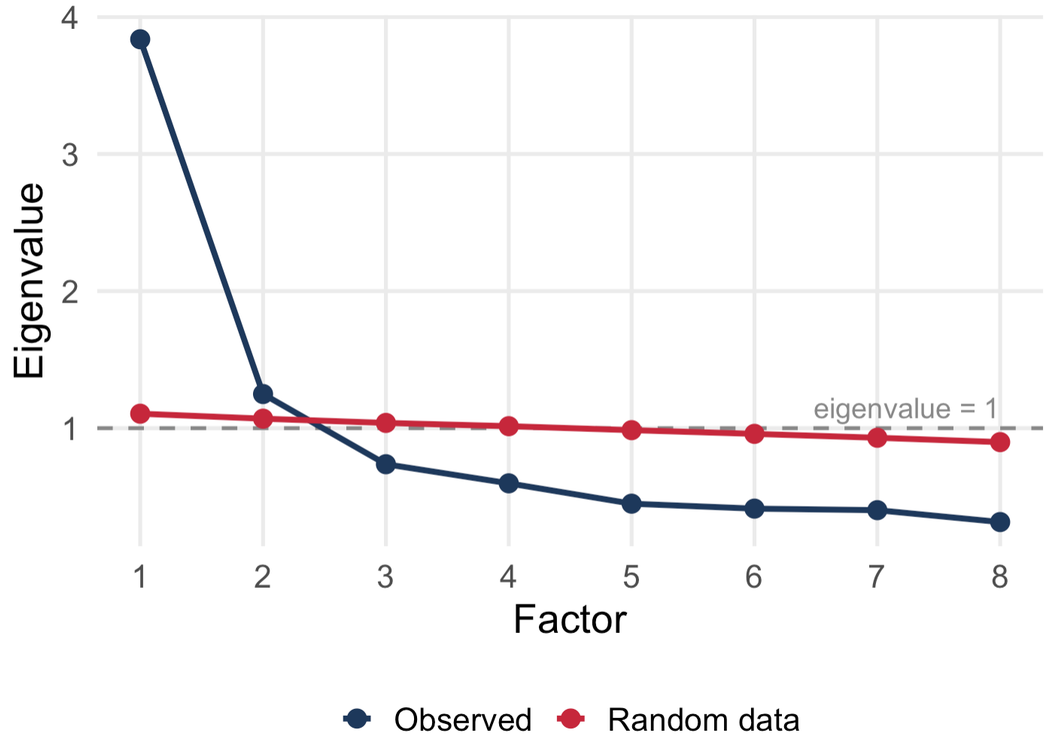
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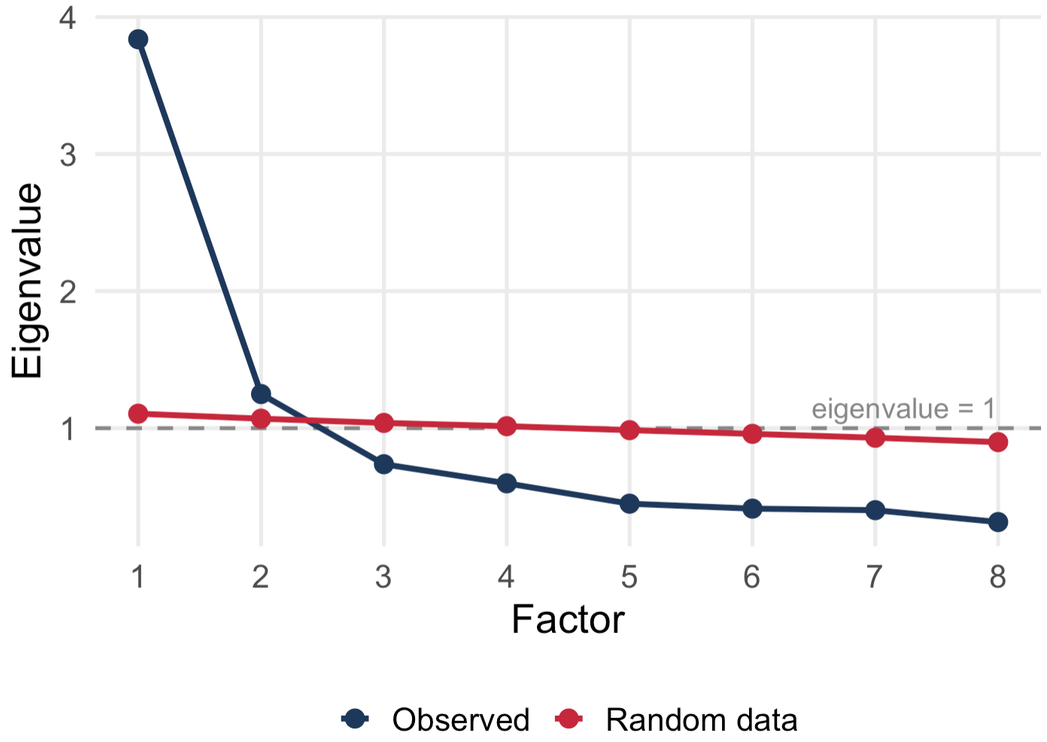
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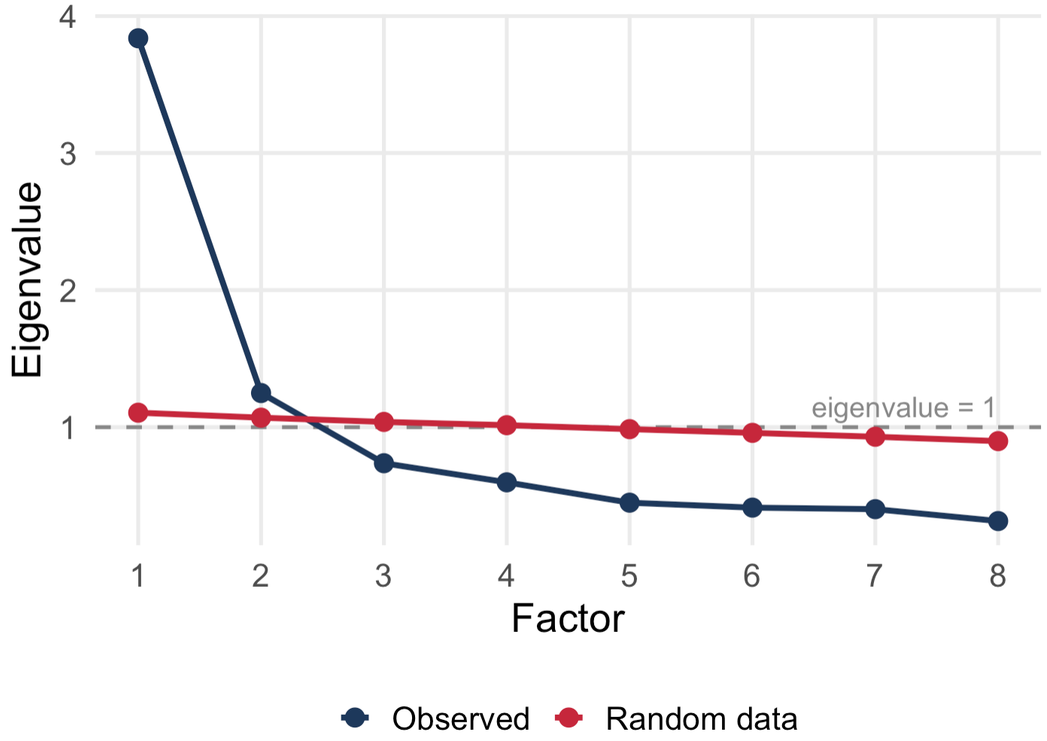
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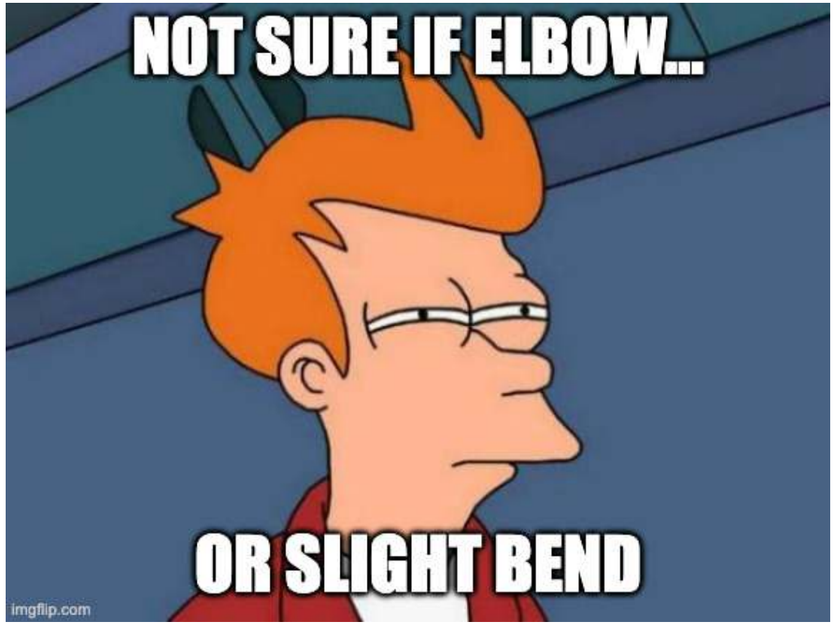
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- **Kaiser**: keep eigenvalues > 1 (worth more than one item). **Scree**: find the "elbow". **Parallel analysis**: beat random data of the same size.

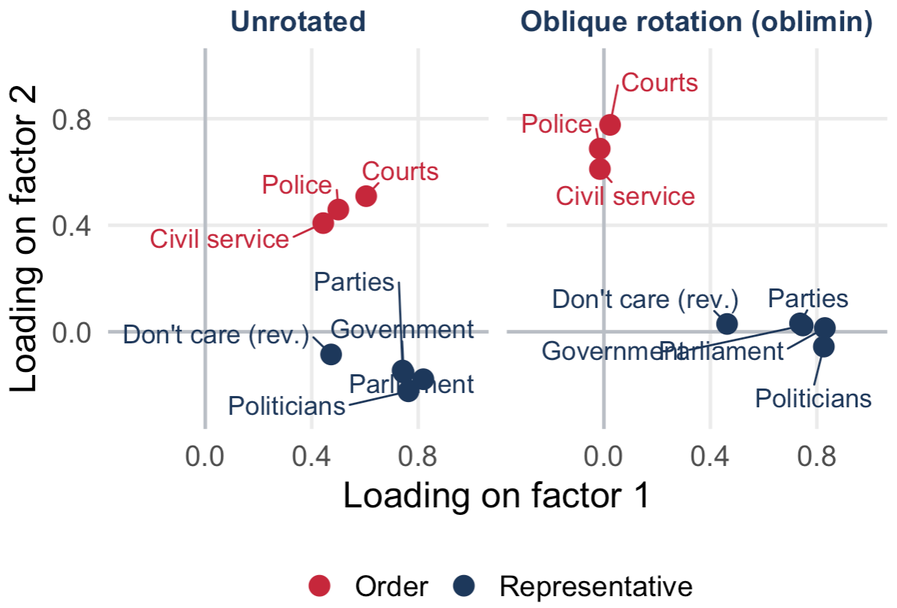


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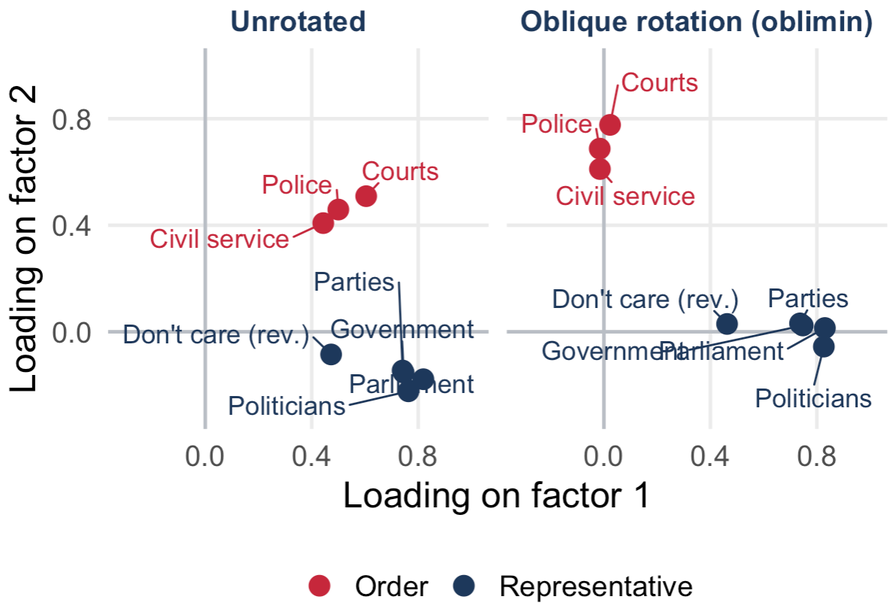
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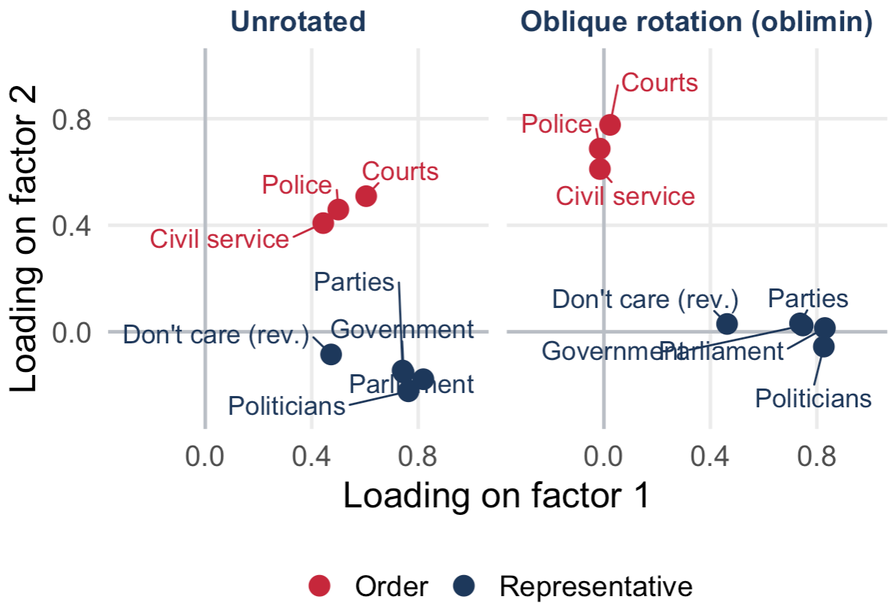
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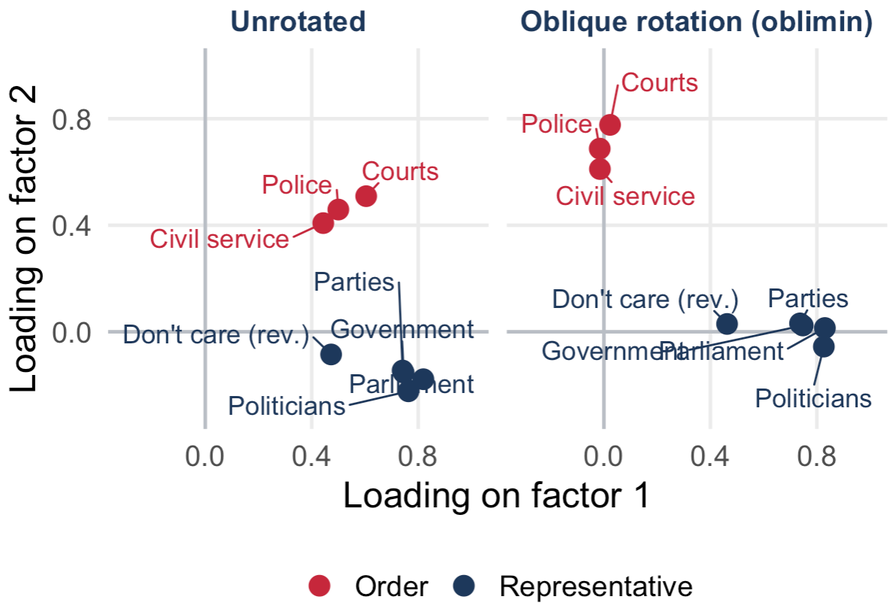
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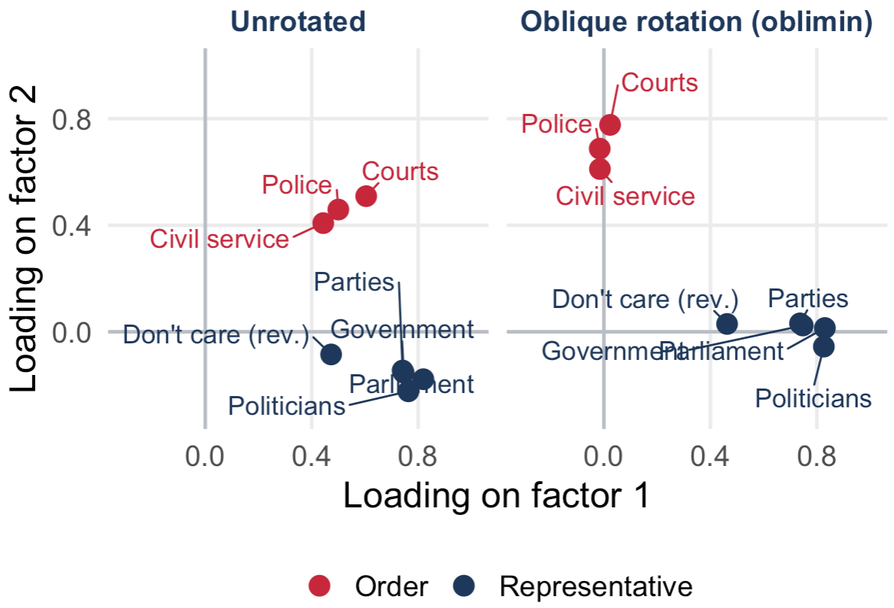
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So why rotate? On the left, every item loads on factor 1, so the factors have no meaning. On the right, representative items sit on one axis and order items on the other: now we can **name** the factors. Same fit, clearer story.

Step 3: rotation in practice

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Do people always rotate?

With **one** factor there is nothing to rotate. With two or more in an **EFA**, almost always: most software rotates by default (`psych::fa()` uses oblimin). A **CFA** never rotates: the loadings you fix at zero already pin the axes down.

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An algorithm searches over all ways of turning the axes for the one that makes the loading table "simplest": big loadings big, small loadings close to zero. You pick the criterion; the software does the search.

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Orthogonal (varimax): the axes stay at 90°, so the factors are forced to be uncorrelated.
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- Rotation changes interpretation, not fit: one more choice to report.

Step 4: read the loading table

	Representative	Order
Parliament	0.83	0.01
Politicians	0.83	-0.06
Parties	0.74	0.03
Government	0.75	0.02
Don't care (rev.)	0.46	0.03
Courts	0.02	0.78
Police	-0.02	0.69
Civil service	-0.01	0.61

Loadings after oblimin rotation (maximum likelihood) on the simulated trust battery. Factor correlation: 0.58.

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- Read by rows: does each item load clearly on **one** factor?
- Watch for **cross-loading** items (two factors) and **weak** items (no factor).

	Representative	Order
Parliament	0.83	0.01
Politicians	0.83	-0.06
Parties	0.74	0.03
Government	0.75	0.02
Don't care (rev.)	0.46	0.03
Courts	0.02	0.78
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Loadings after oblimin rotation (maximum likelihood) on the simulated trust battery. Factor correlation: 0.58.

Step 4: read the loading table

- Read by rows: does each item load clearly on **one** factor?
- Watch for **cross-loading** items (two factors) and **weak** items (no factor).
- The police item is not weak: it is a **good indicator of a second factor**. The reverse-coded item is the weakest.

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Compulsory reading. Read it as a worked example of building a scale: concept → items → EFA → CFA in new samples → IRT short form → reliability → invariance → validation.

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Meet the nine items

Shuffled. Which ones belong together? (Democrats see “Republicans” as the out-party, and vice versa.)

- My identity as a [in-party] is rooted in moral principles.
- [Out-party] live in a different world from us [in-party].
- If I found out a friend of mine was a [out-party], I would want to stop spending time with them.
- I feel as though [in-party] are very different from [out-party].
- My identity as a [in-party] reflects my beliefs about the difference between right and wrong.
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- As a [in-party], I would not want to be friends with someone who was a [out-party].
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Sort them into **three groups** before the next slide.

Three subdimensions, three items each

The Affective Polarization Scale (Campos & Federico 2026, Table 1):

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Notice what **response processes** (Week 4) look like here: every item names **ordinary partisans or one's own identity**, never party elites. That is a design choice against Druckman & Levendusky's problem.

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- Exactly the four steps, with the rules written down: thresholds (.40, 75%) and a fixed number of factors.
- Count the decisions. Each one is defensible, and each one is a path that could have gone another way. That is why the next step uses **new samples**.

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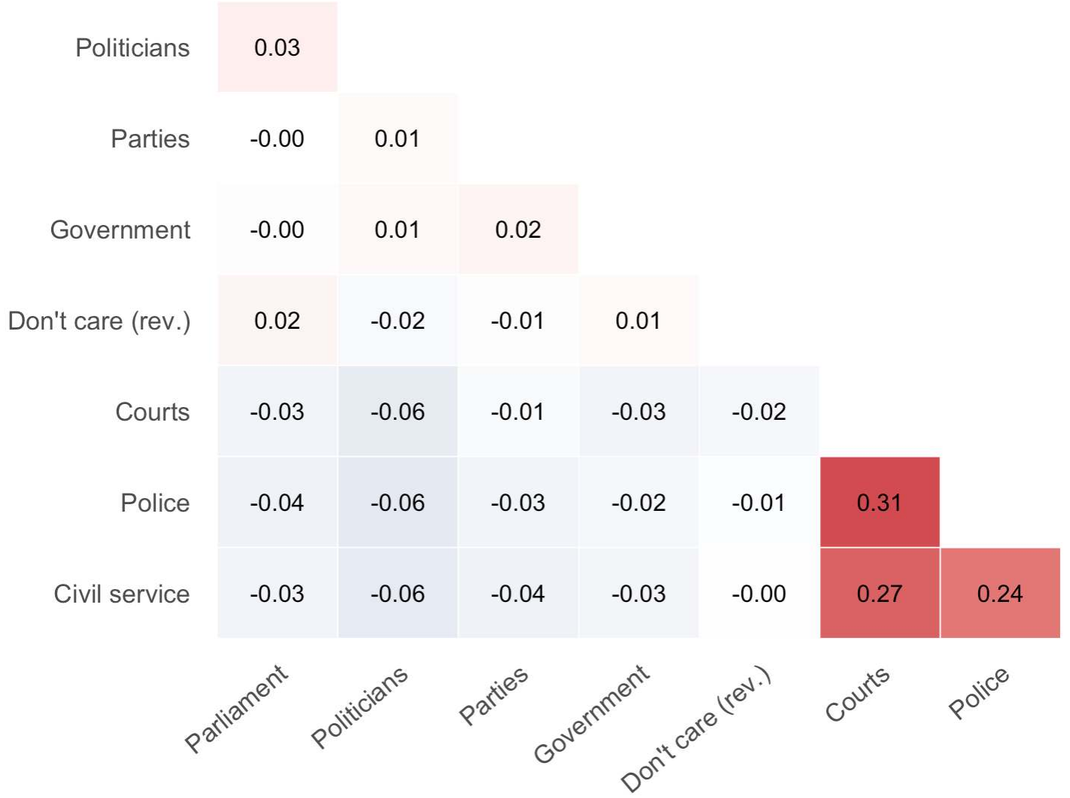
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- **Fit** = how close the predicted correlations are to the observed ones. Every missing arrow is a claim, as in a DAG, and the data can reject it.

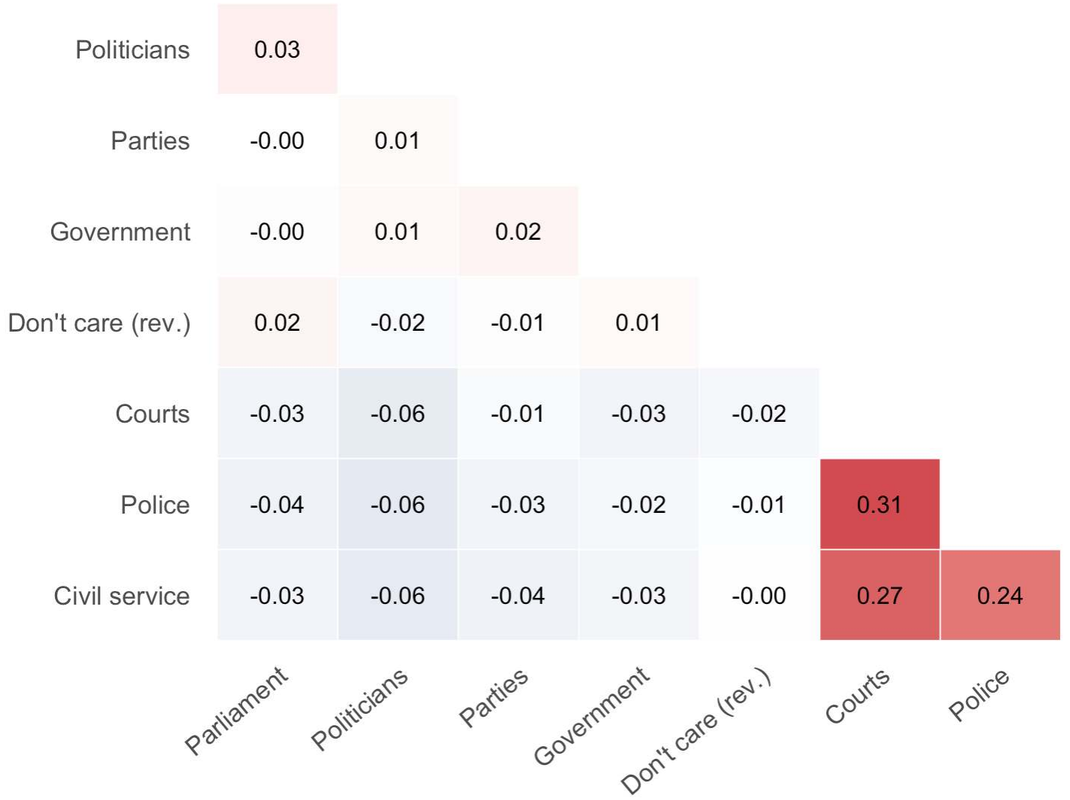
Where does “one form of trust” fail?



Residual correlations (observed minus implied) from the one-factor CFA. Red cells: pairs the model under-predicts.

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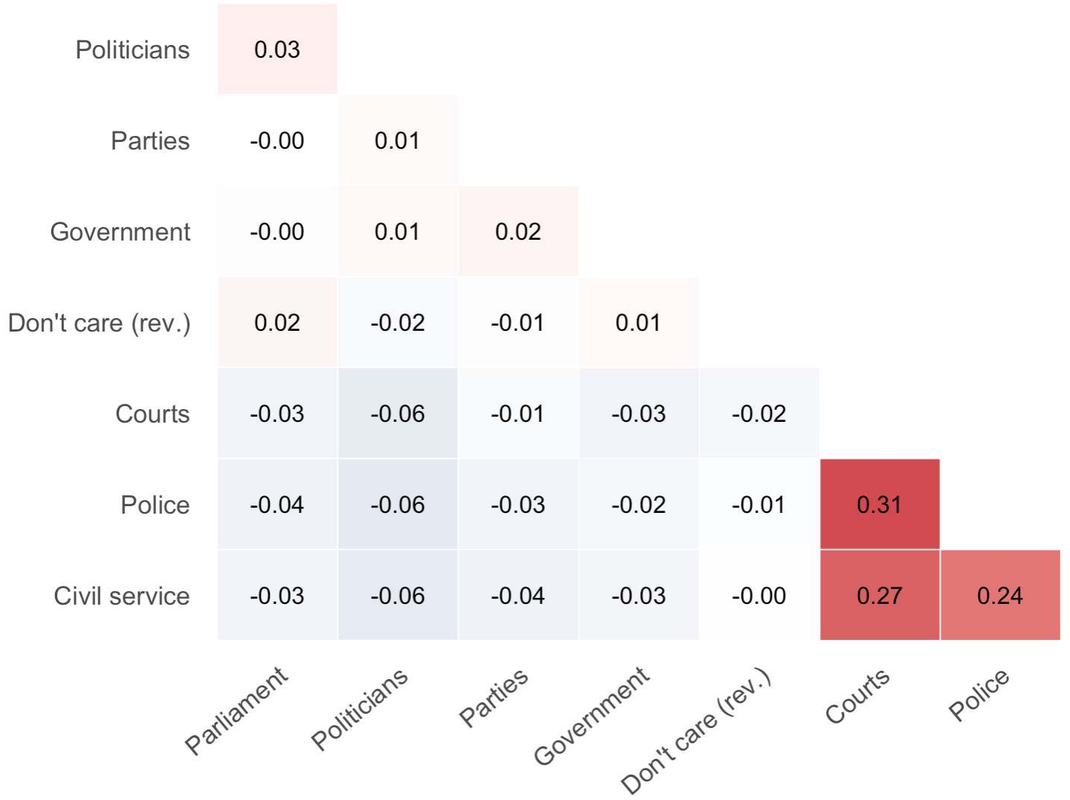
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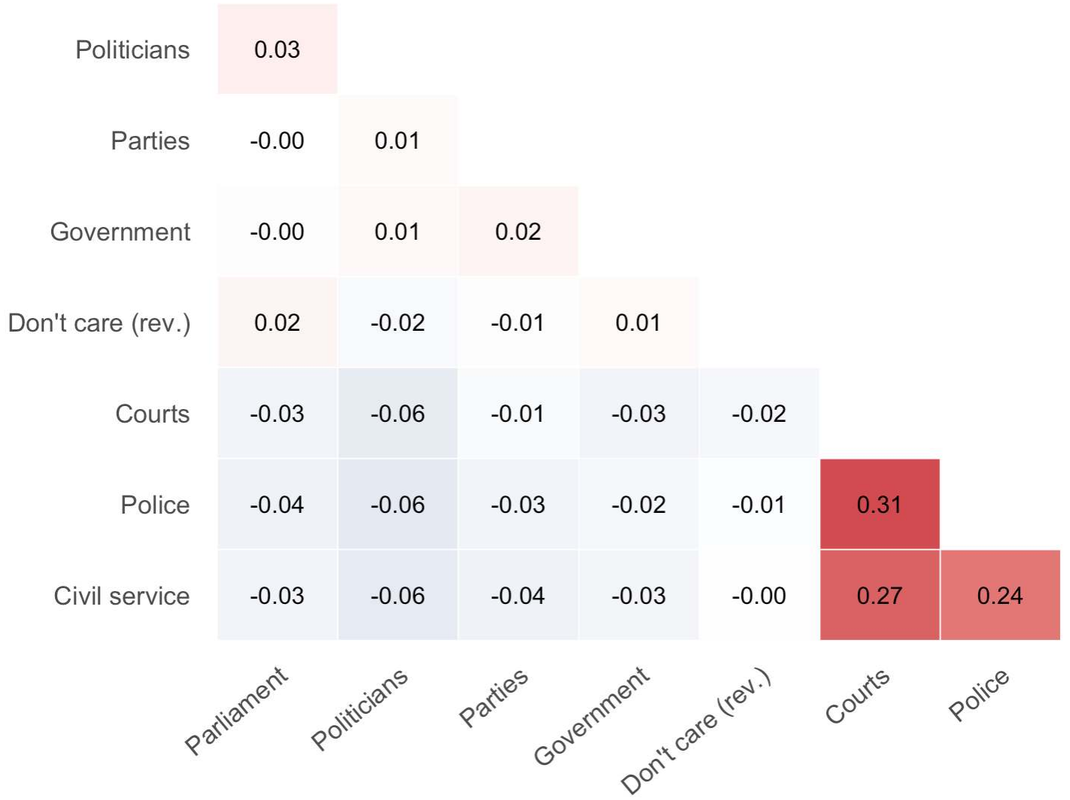
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- Fit the **one-factor** CFA and compare its implied correlations with the observed ones.
- The residuals are near zero almost everywhere, except among **courts, police, and the civil service**.
- The model cannot explain why those three hang together more than “trust” says they should. That pattern **is** the second factor.



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Comparing the two theories

```
one <- "trust =~ parliament + politicians + parties + government + dont_care_r + courts + police + civil_service"
two <- "representative =~ parliament + politicians + parties + government + dont_care_r
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- The two-factor model fits much better, and the factors correlate at 0.59: **related, but distinct**. In real data this is an empirical question, and the answer can differ across countries (end of today).

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Cutoffs are conventions, not tests. A model that fits is not proven true; compare **theoretically motivated alternatives**, and look at the residuals.

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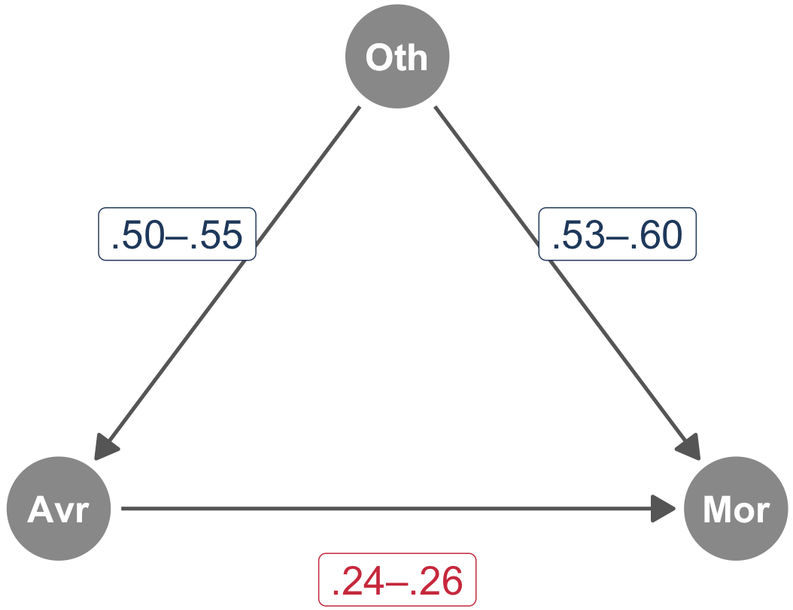
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- Same move as our trust example: **several theories, several models, one comparison**, on data that did not build the scale.

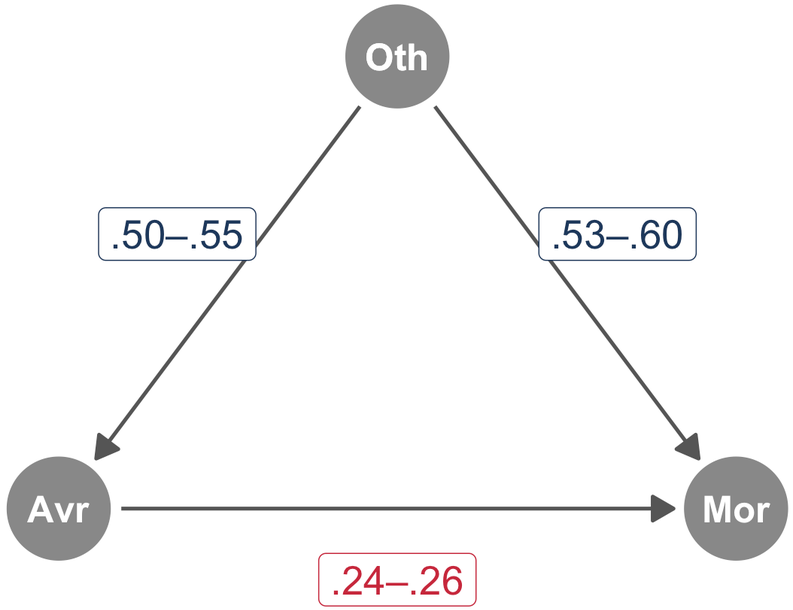
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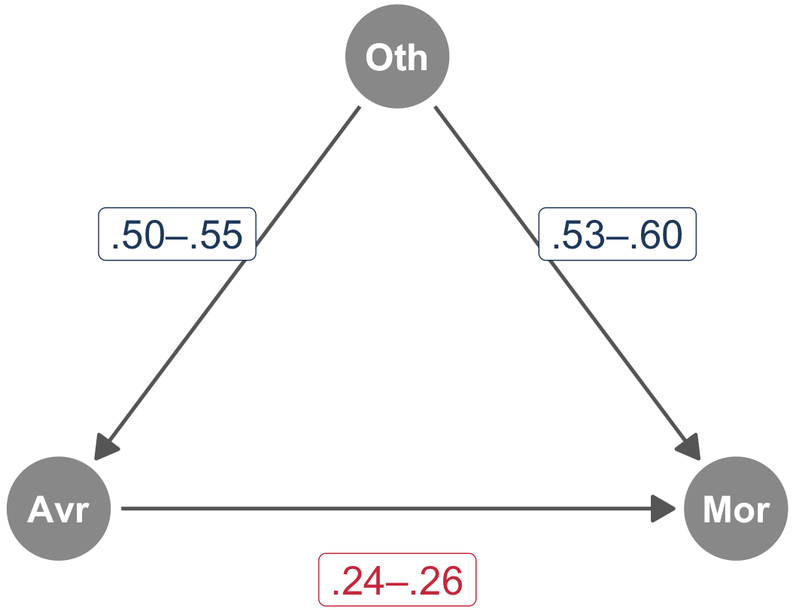
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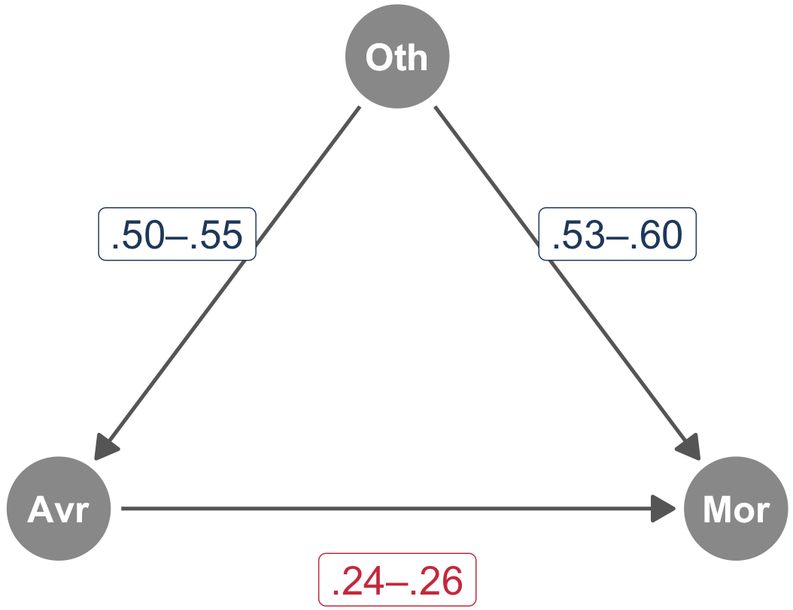
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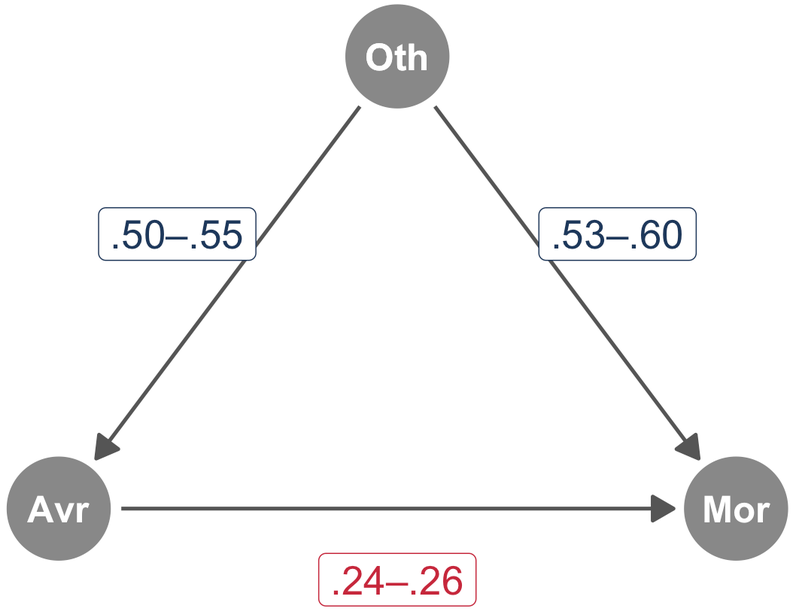
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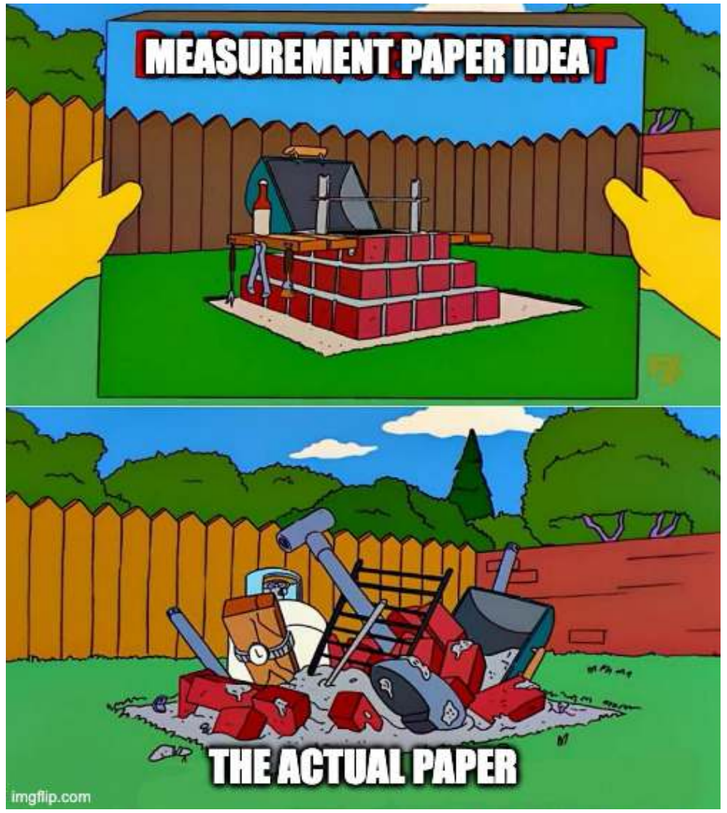
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- Report what you dropped and why, with the full-item results in an appendix.

The forking paths of scale building

- Building a scale involves many decisions: which items, how many factors, which rotation, which items to drop, whether to merge factors.
- Made **after seeing the data**, each one is a **forking path** (Week 1) and a questionable measurement practice (Flake & Fried, Week 4).

> Two protections: **pre-register** the intended structure, or **split the samples**: explore on one, confirm on another. Campos & Federico do both (Study 3 is preregistered; Studies 2–4 confirm Study 1).

- Report what you dropped and why, with the full-item results in an appendix.



Running the EFA for the seventh time, with a different rotation.

Your turn ✏️ (~2 min, as a group)

A new battery on **perceived cycling safety**: an illustrative EFA loading table (simulated data).

Item	Factor A	Factor B	Factor C
Safe next to buses	0.78	-0.02	0.00
Safe at junctions	0.80	-0.10	0.00
Safe in fast traffic	0.66	0.31	0.06
Safe cycling at night	0.03	0.76	0.02
Safe from theft	-0.01	0.73	-0.02
Safe from harassment	-0.01	0.69	0.04
Lanes are continuous	-0.03	-0.01	0.74
Lanes are wide enough	0.04	-0.01	0.66
Enough bike parking	-0.06	0.09	0.15

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What do we see here?

Break

10 minutes

Item response theory

Why IRT? Not just “factor analysis for yes/no”

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People and items on one scale

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- Factor analysis models **covariances** between items. IRT models the **probability of each answer** for each person: a regression of every answer on an unobserved predictor, item by item.

Think of an exam

IRT grew up in educational testing. Its vocabulary is easiest to learn there.

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- For attitudes, read “ability” as the **level of the trait** (trust, polarization) and “right answer” as **agreeing**. The math is the same.

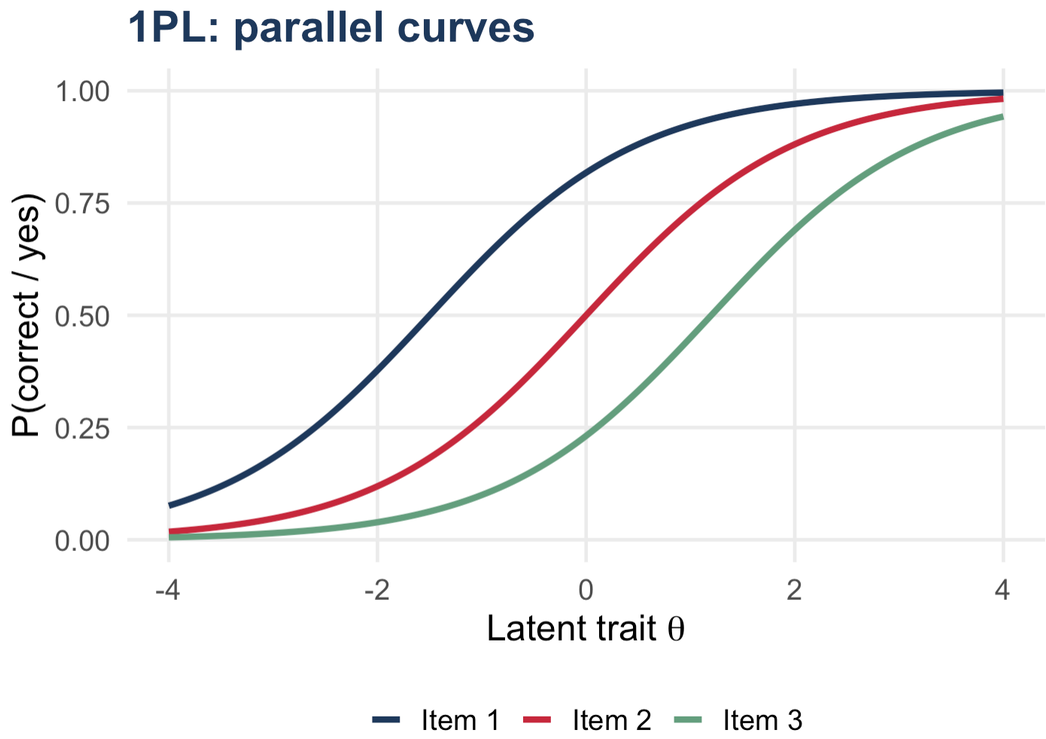
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Three illustrative items, same difficulties in every panel. Each model adds one parameter per item.

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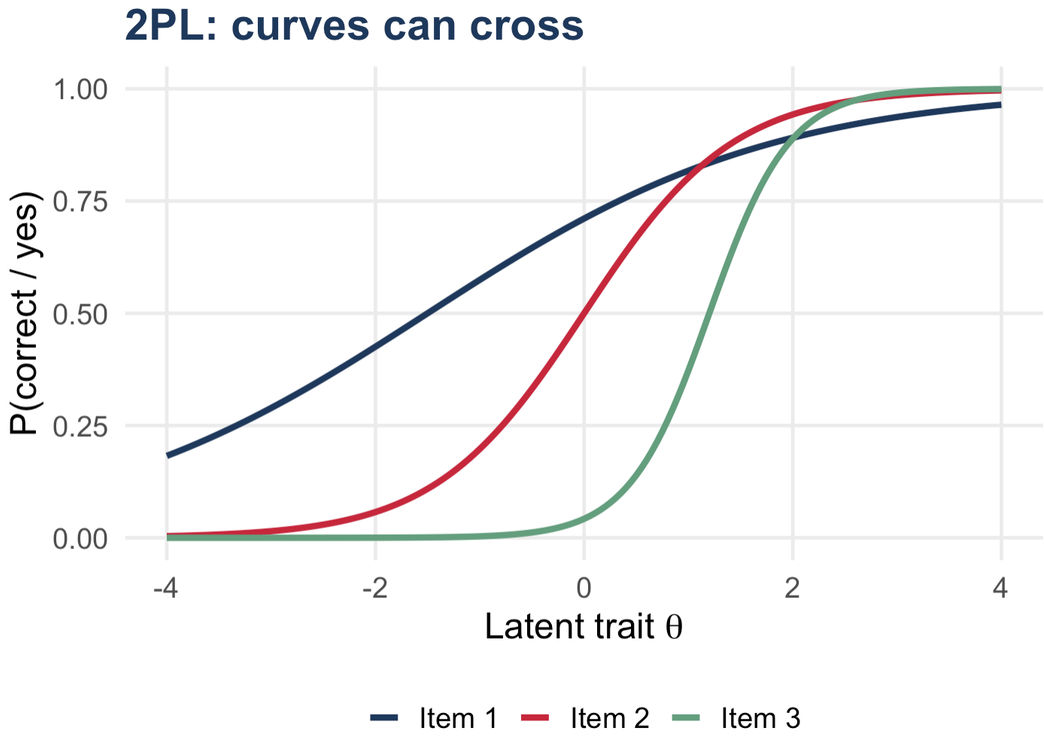
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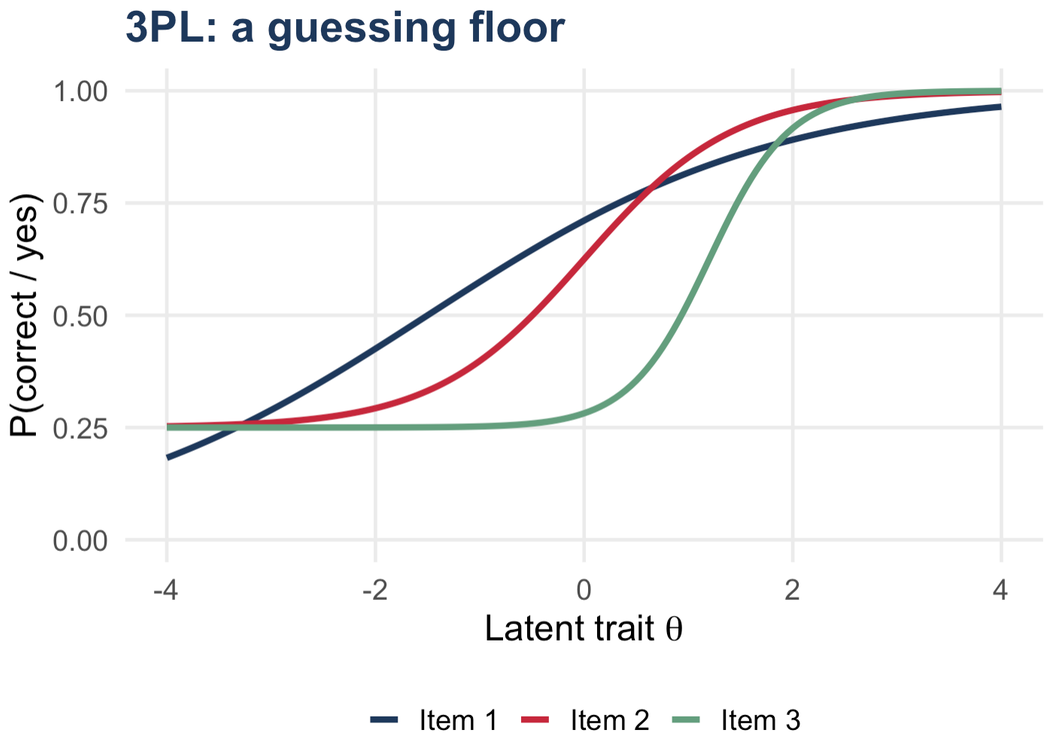
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3PL: a lower floor c_j for **guessing**.

$$P(y_{ij} = 1) = c_j + (1 - c_j)\,\Lambda\big(a_j(\theta_i - b_j)\big)$$

where θ_i is person i's latent position and Λ the logistic function (next week: the logit link)



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From my own research: the INSPIRE inventory

Measuring scientific evidence consumption literacy for public policy with the INSPIRE inventory

Sebastian Ramirez-Ruiz (European University Institute)*
Simon Munzert (Hertie School)

Abstract. Making sense of scientific evidence is critical for effective public policy, enabling policymakers and the public to make informed decisions on complex issues. However, existing measures may not fully address the specific demands of policy contexts. We present the Inventory for Numeracy, Statistics, and Policy-oriented Inference and REasoning (INSPIRE), a 30-item instrument assessing scientific evidence consumption literacy for policy. INSPIRE integrates four subdomains: statistical reasoning, data and visualization literacy, causal inference, and understanding of the scientific method. The instrument was developed and validated using Item Response Theory across four samples: a general online population, policy students in a training program, policy professionals, and participants in a forecasting study. Beyond strong psychometric properties, scores predict performance on applied reasoning tasks, ability to reject inaccurate scientific beliefs on contested issues, and accuracy in interpreting quantitative evidence. We provide the validated item bank alongside guidelines for flexible deployment in research and applied settings.

Keywords. scientific evidence consumption | scientific reasoning | public policy | evidence-informed policymaking | item response theory (IRT) | Test instrument

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- Next: four of its items, and what the parameters say about them.

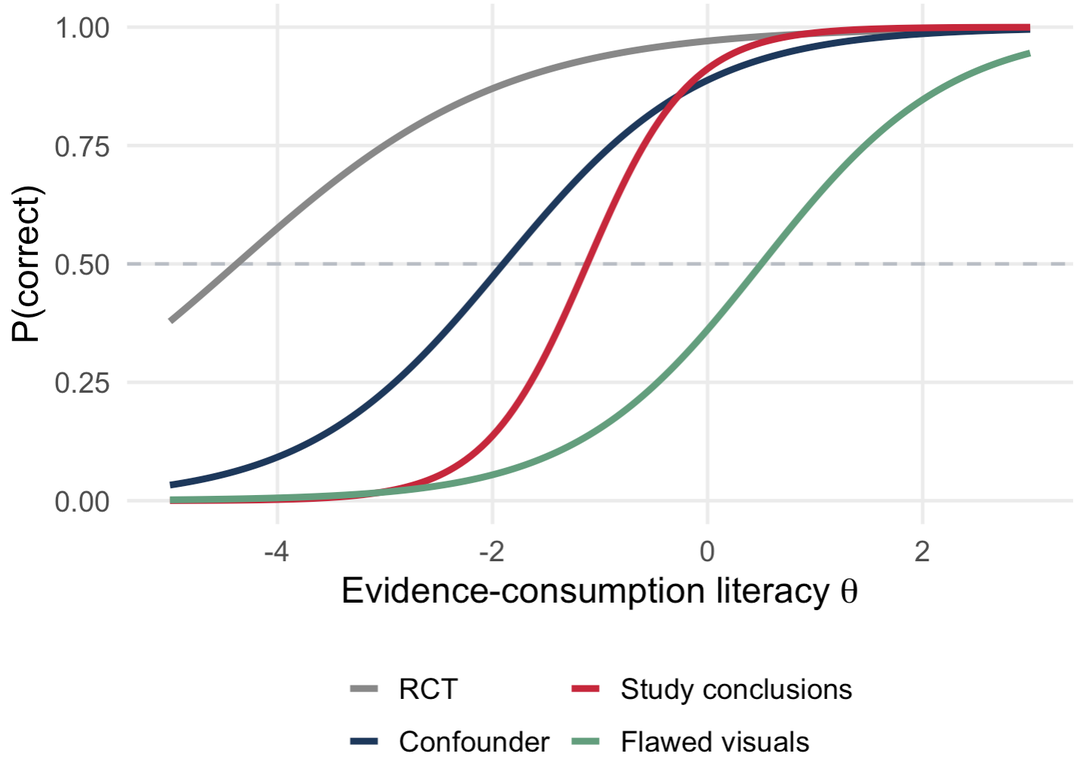
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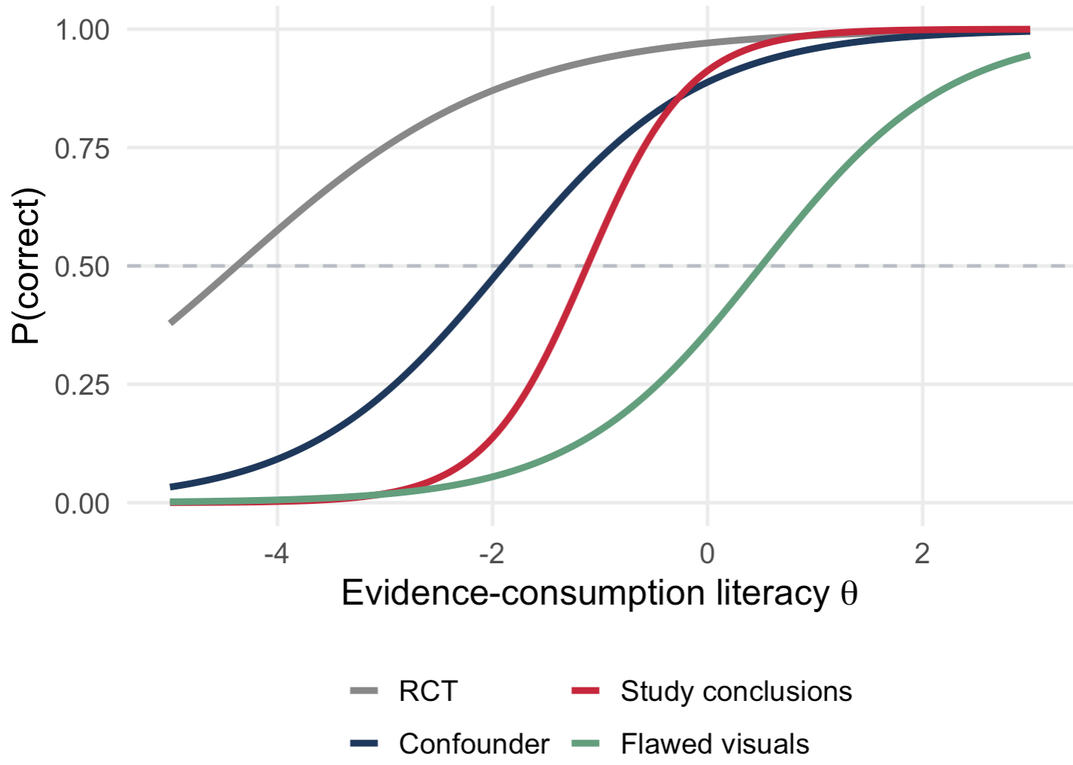
Four real test items



2PL curves for four items of the INSPIRE inventory, using the published estimates (N = 470). Where each curve crosses the dashed line is its difficulty; its steepness is its discrimination.

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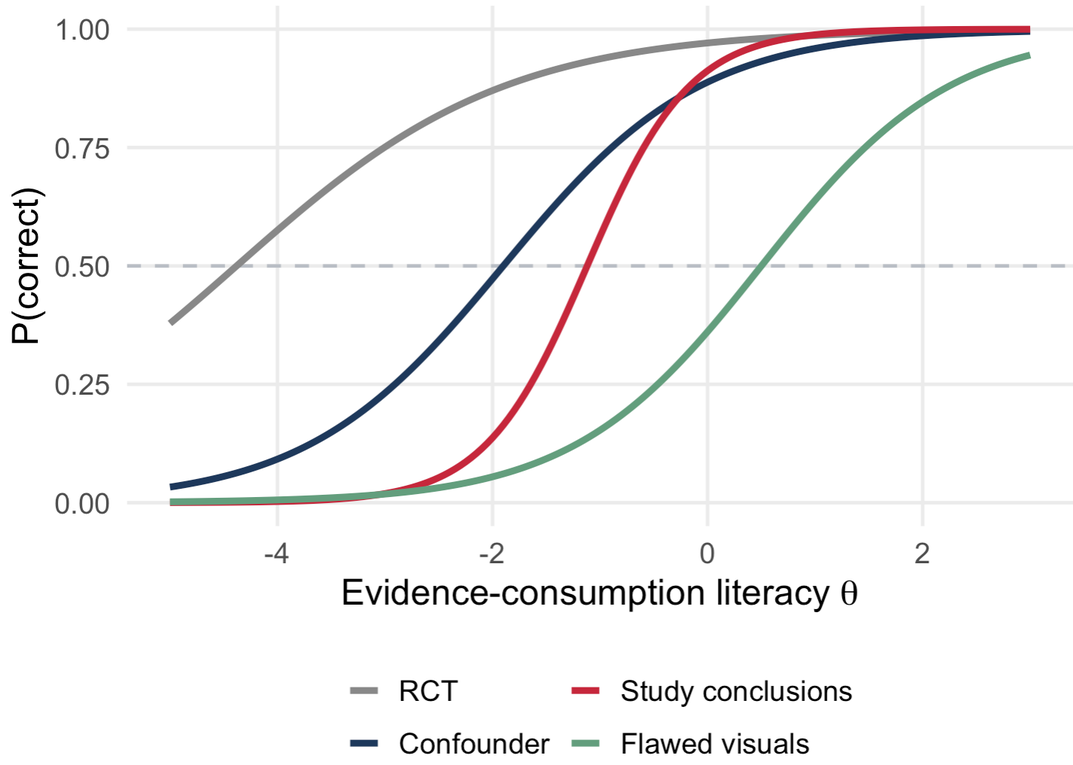
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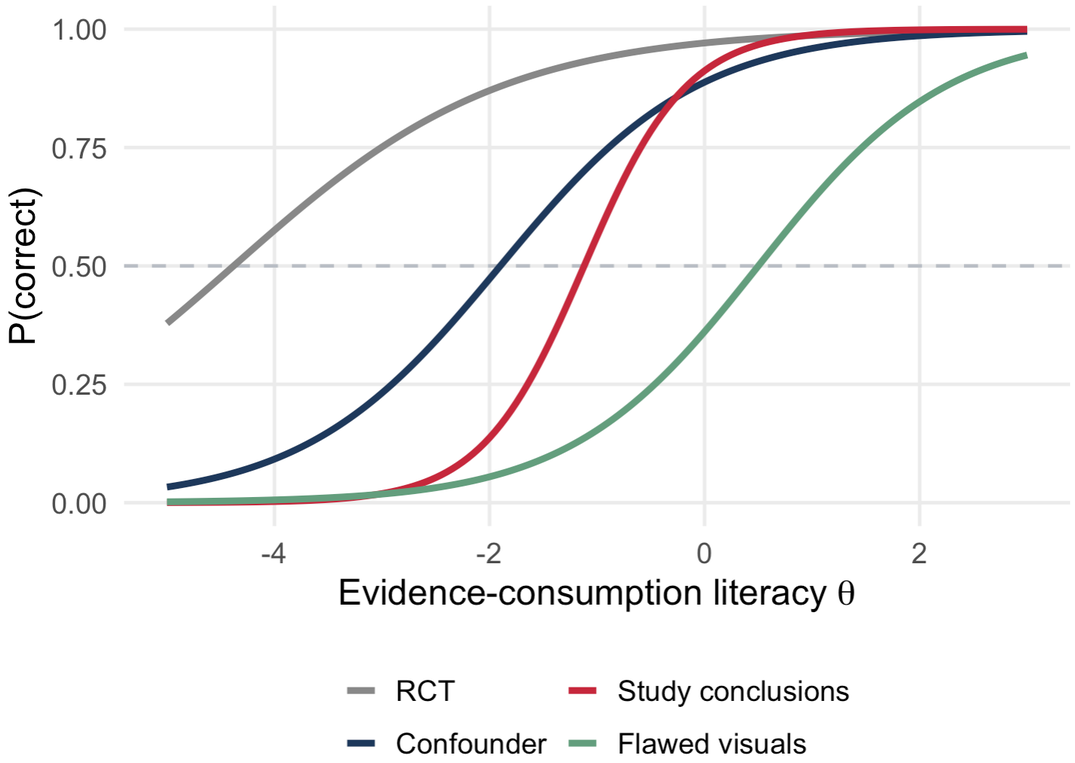
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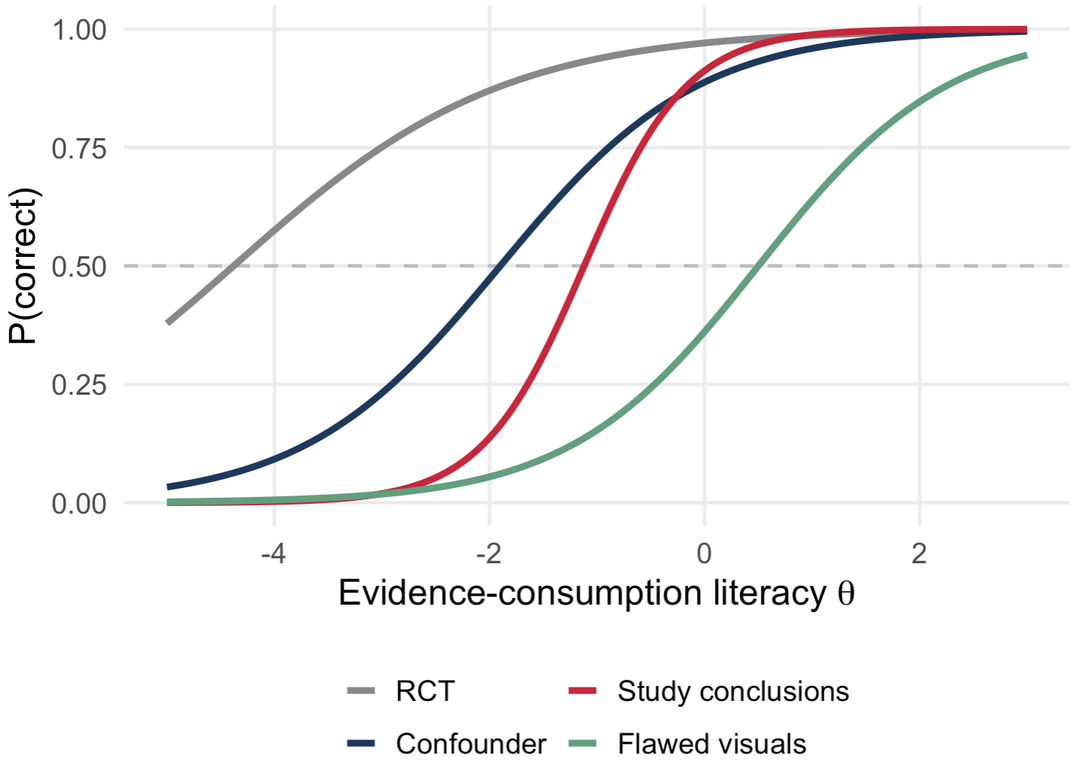
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- **Flawed visuals:** spot what is *not* a problem in a misleading graph. a = 1.14, b = **0.50**. The hardest of the four.



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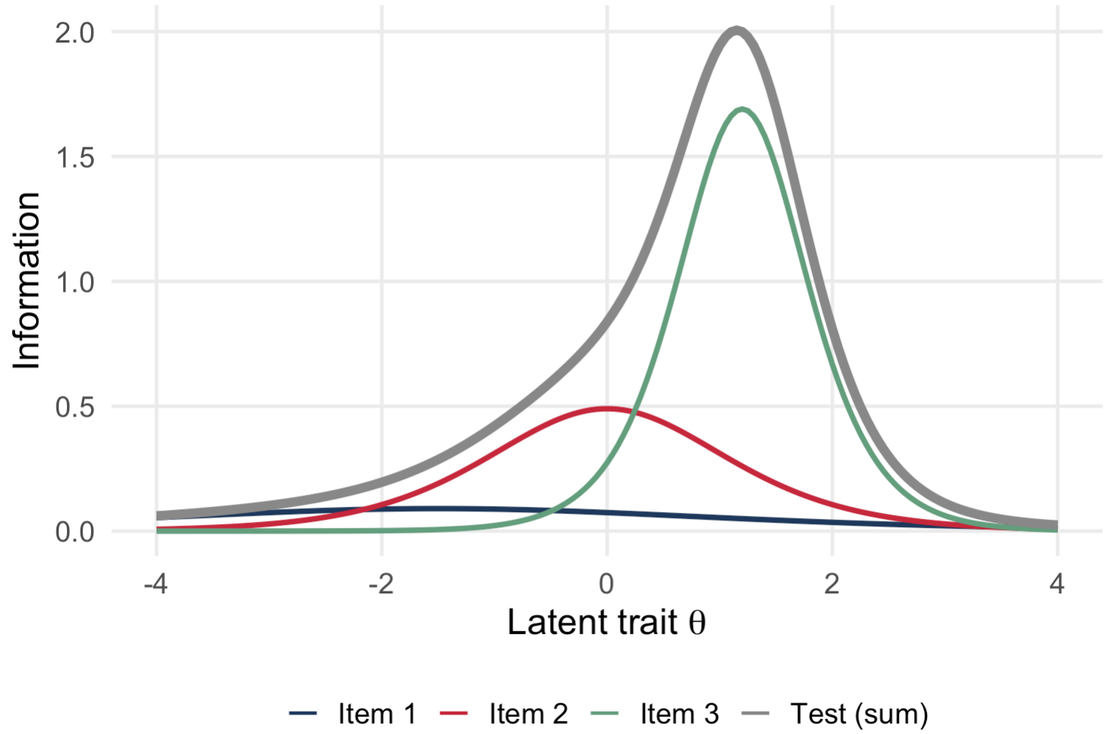
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- Choosing among them is like choosing among CFAs: **theory first**, then compare fit. In our scale, items are mostly multiple choice, yet the 3PL did **not** beat the 2PL: the format suggests a model, the data still get a say.

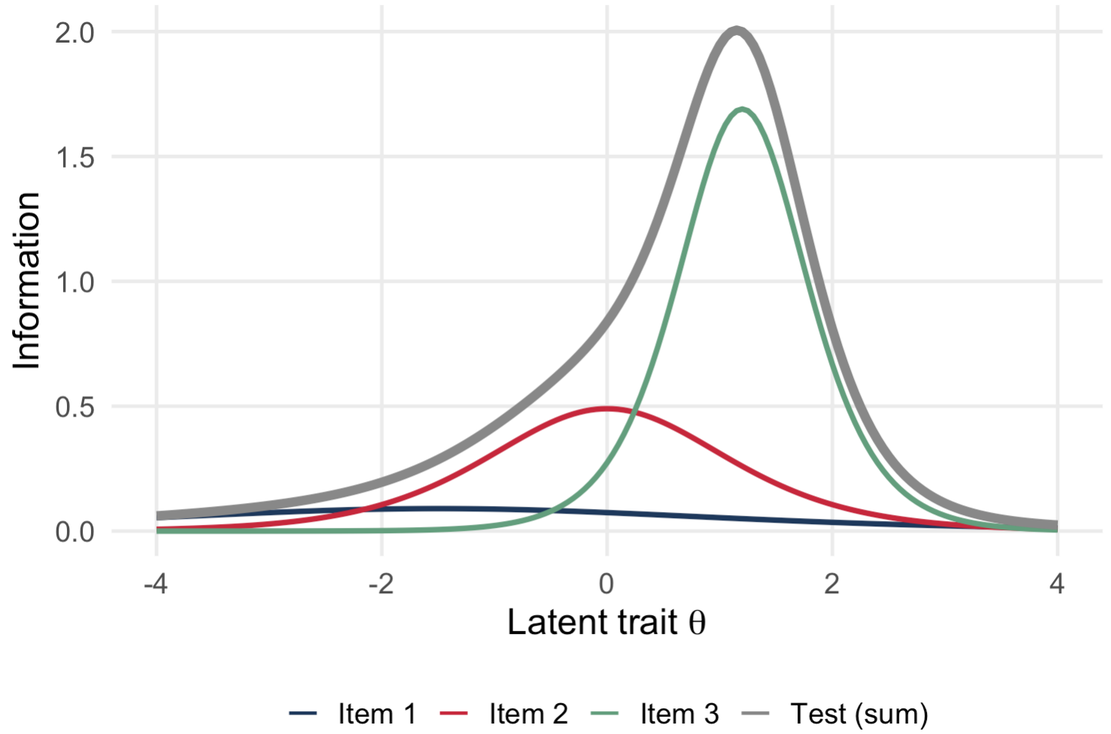
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The three 2PL items from before. The flat item adds little anywhere; the steep one is very informative, but only near its own difficulty.

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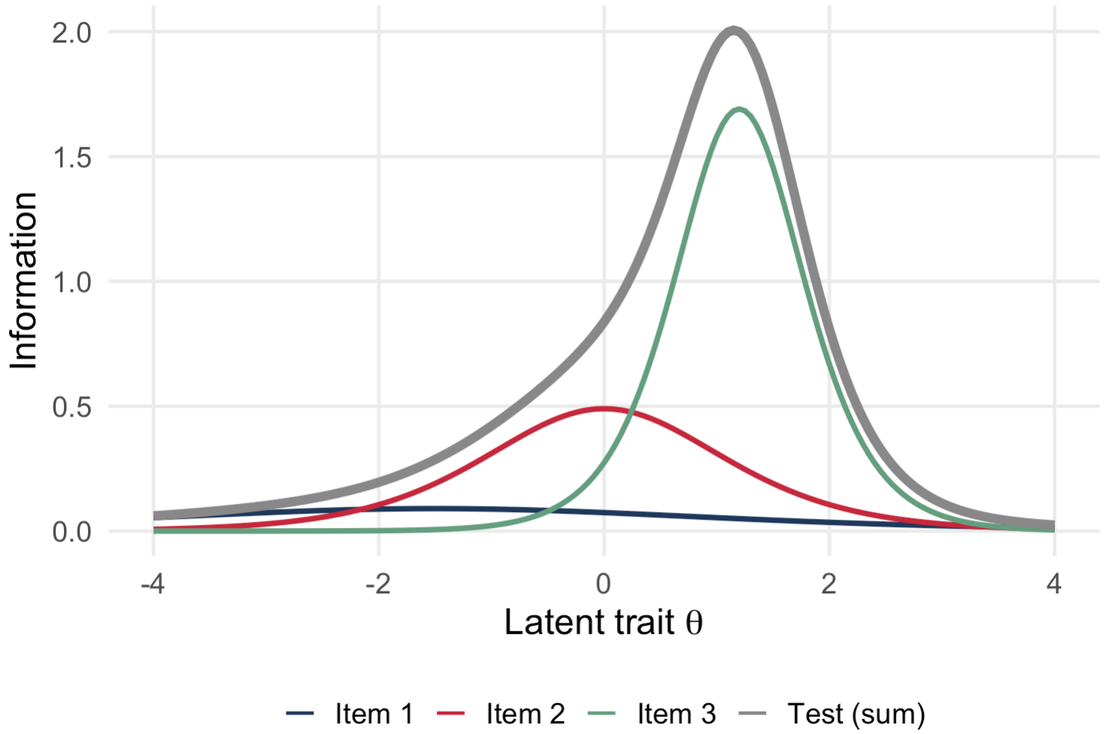
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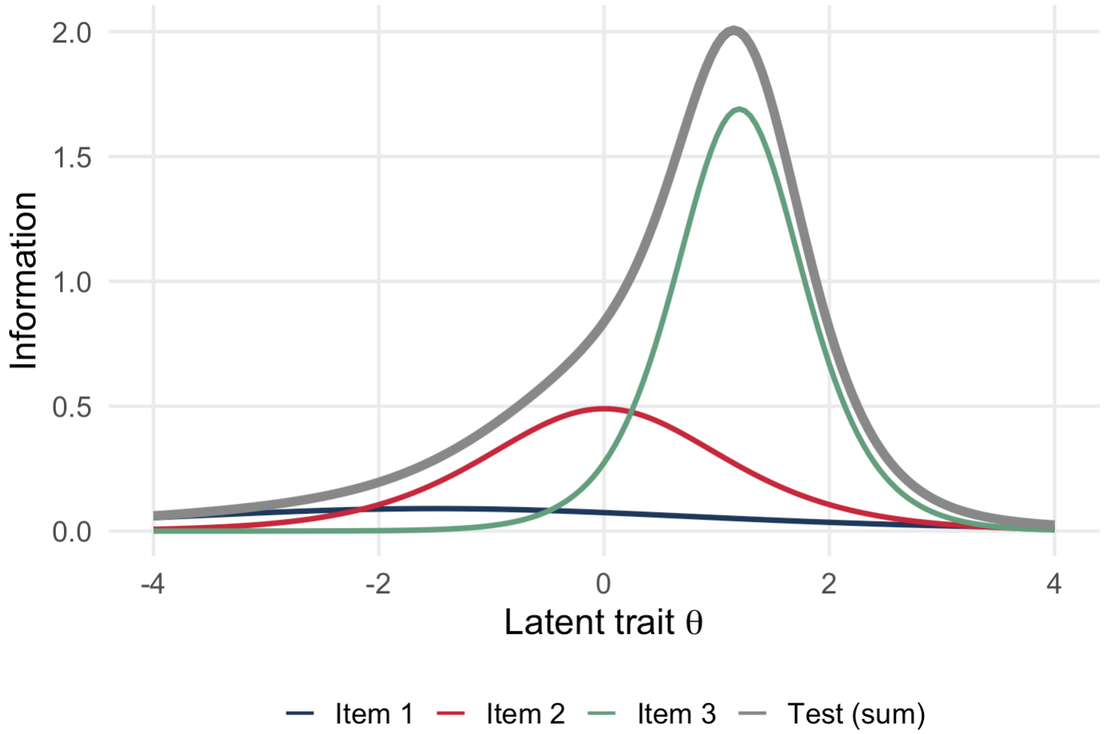
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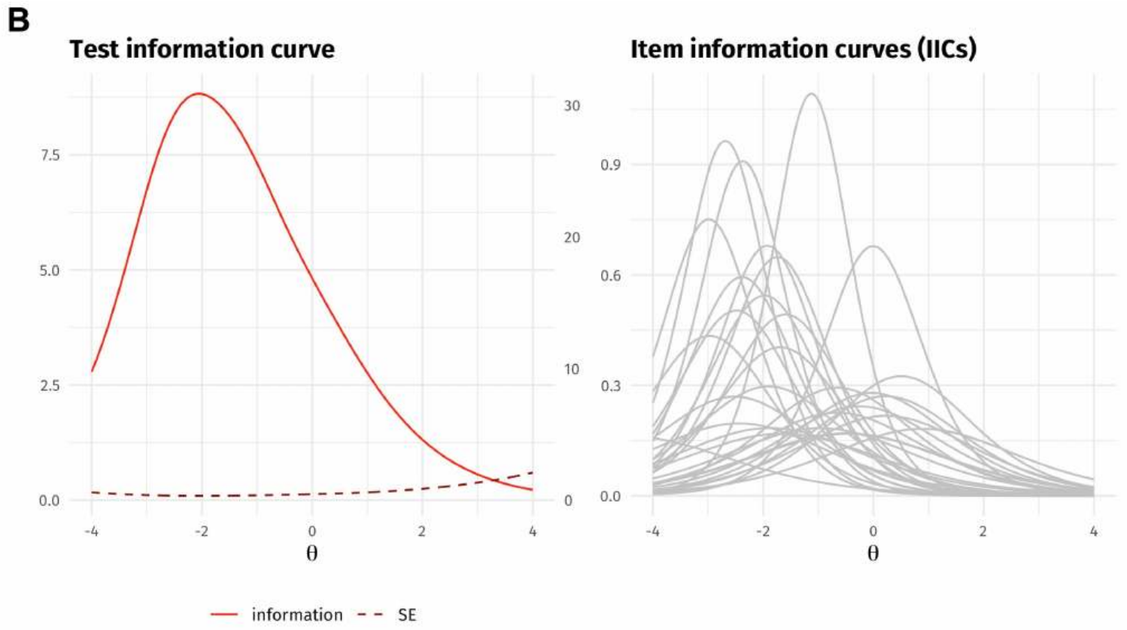
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- A scale **can be precise for some people and blind for others**: no single reliability number shows that.



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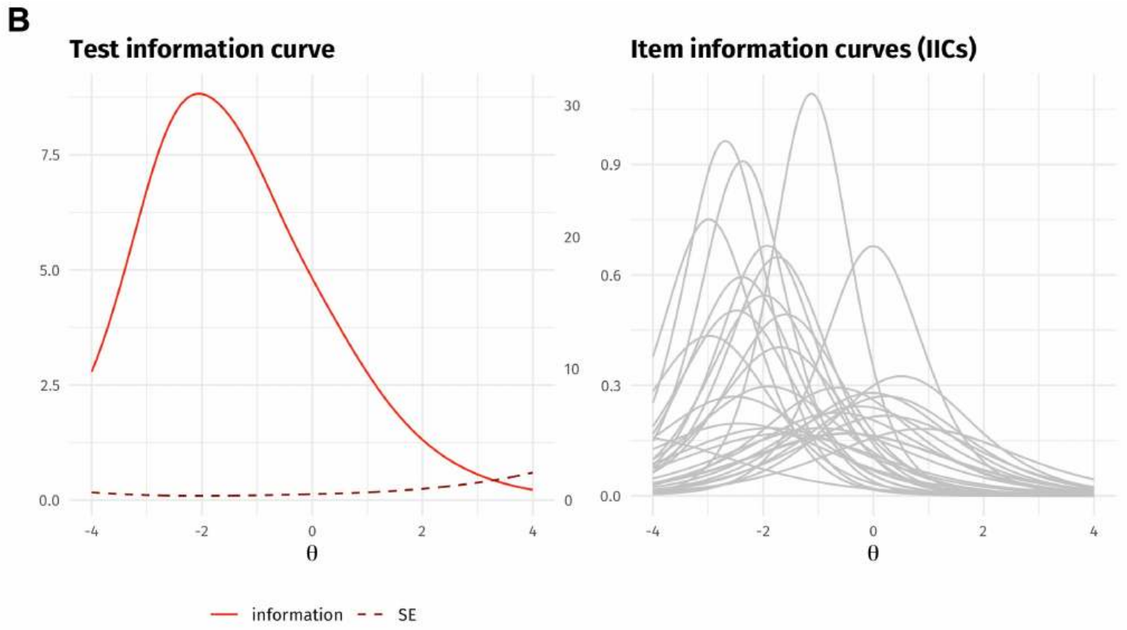
Precise where it matters for policy



INSPIRE, 30 items, 2PL. Left: test information (red) and its standard error (dashed). Right: each item's information. Ramirez-Ruiz & Munzert, Figure 1B.

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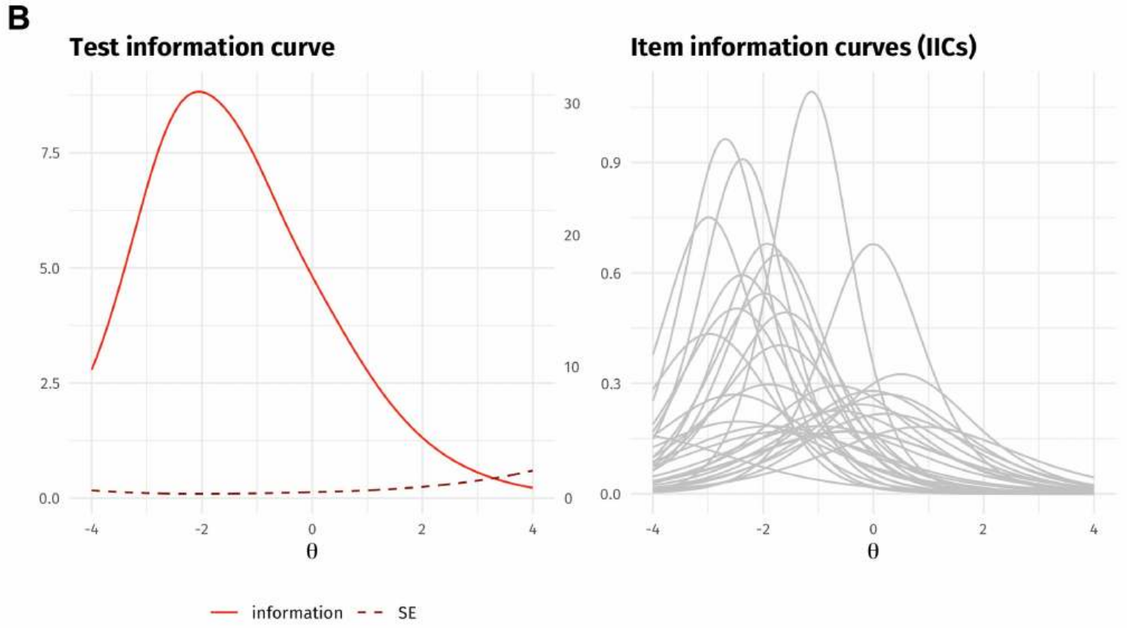
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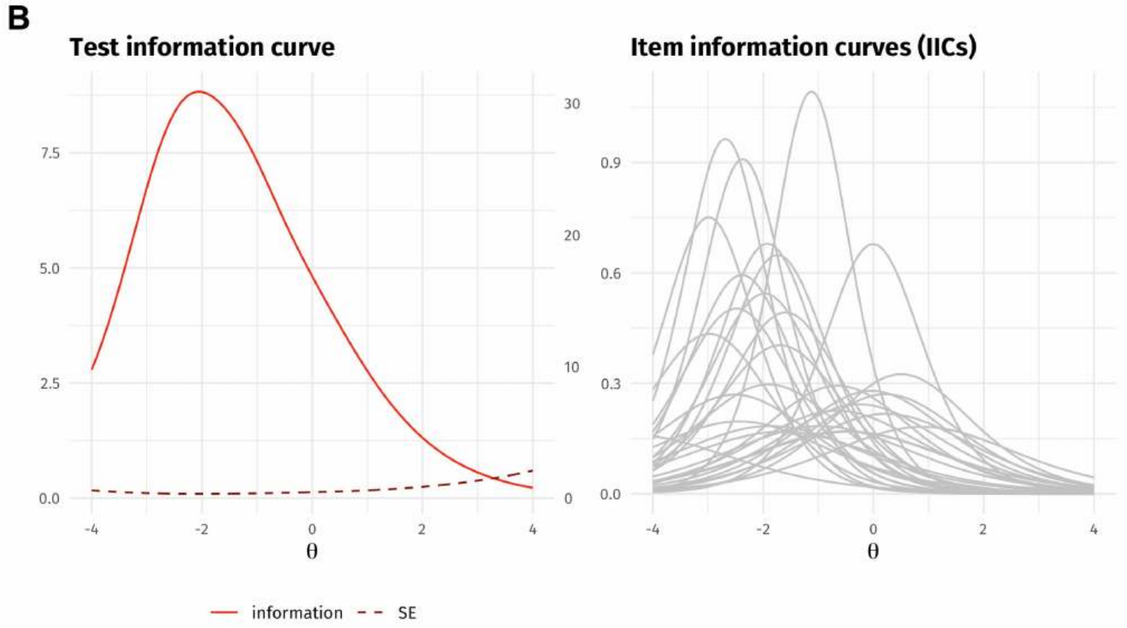
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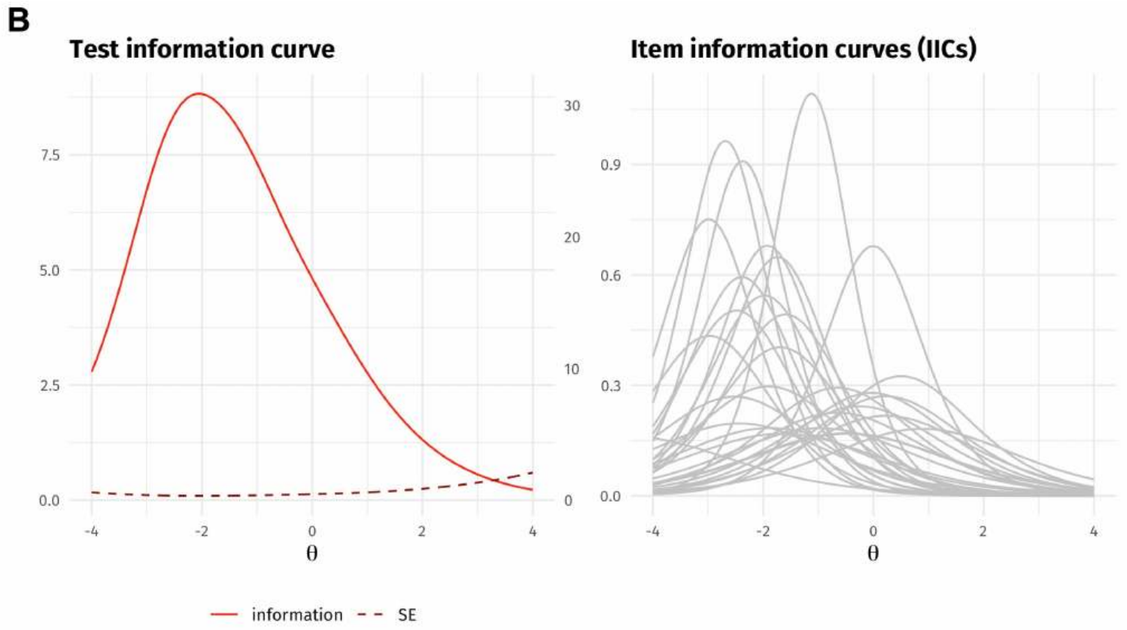
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- It is most precise for the people who most need support. To distinguish among the strongest consumers of evidence, we would need harder items, and the peak would move right.
- The purpose and the population, not a reliability number, **decide where a measure should be precise**.



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Pruning items in two published scales

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Affective polarization: 20 → 9 items

- A graded response model within each subdimension.
- Rank items by the area under their information curve; keep the two best.
- Choose the third for **coverage** of a wider range of θ, face validity, and non-overlap.
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- Two careful teams, two sets of rules, both written down. Notice the **numbers**: 0.7, 0.2, three items per dimension. Where do they come from?

Thresholds are conventions, not laws



Me, building a measurement inventory with Simon.

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- Let **theory** do the work first: what is the construct, for whom, and for what? Then pick rules, write them down **before** looking, and show what happens if they move (.6 instead of .7?).

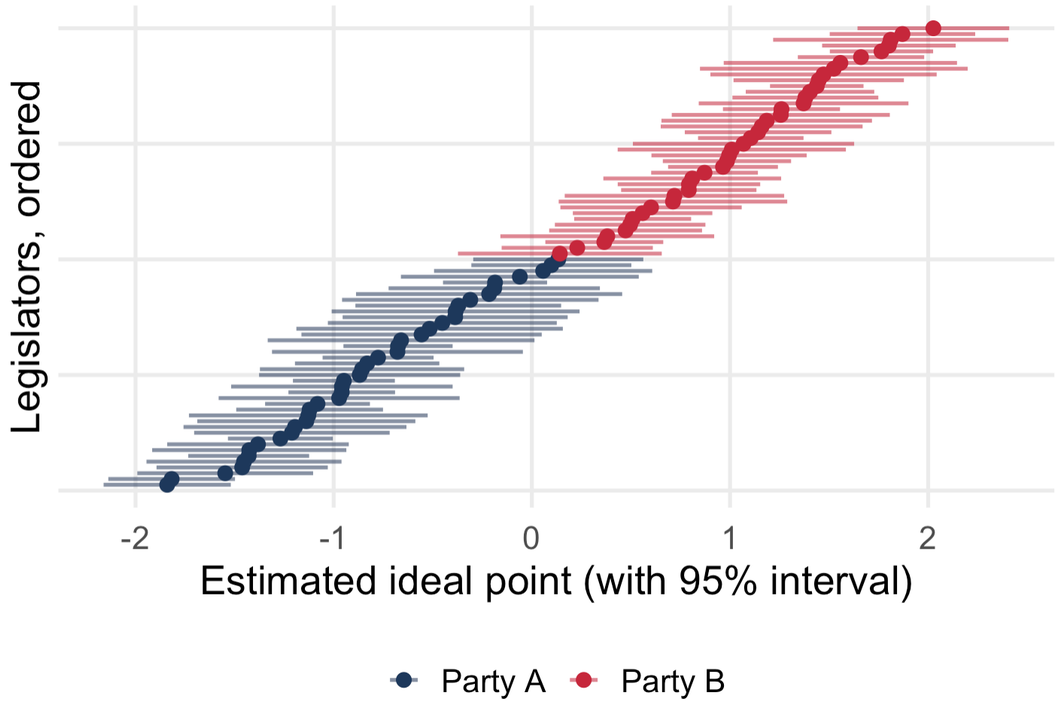


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Ideal points: you may already know this one



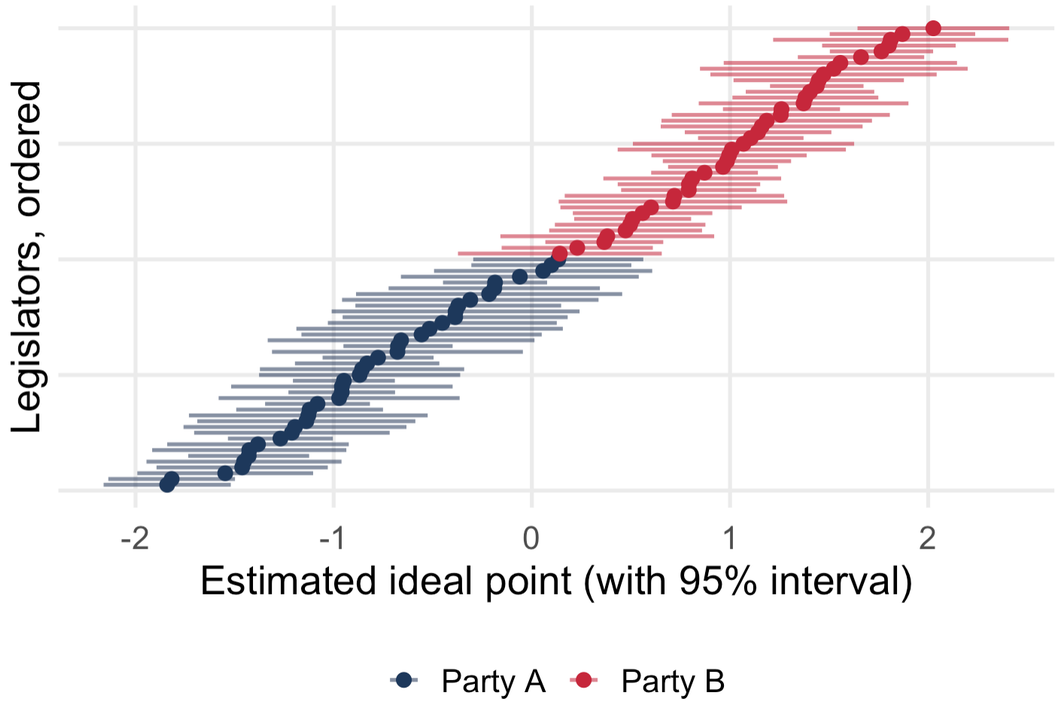
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A simulated legislature of 80 members. Each dot is an estimated ideal point; the bars are its uncertainty.

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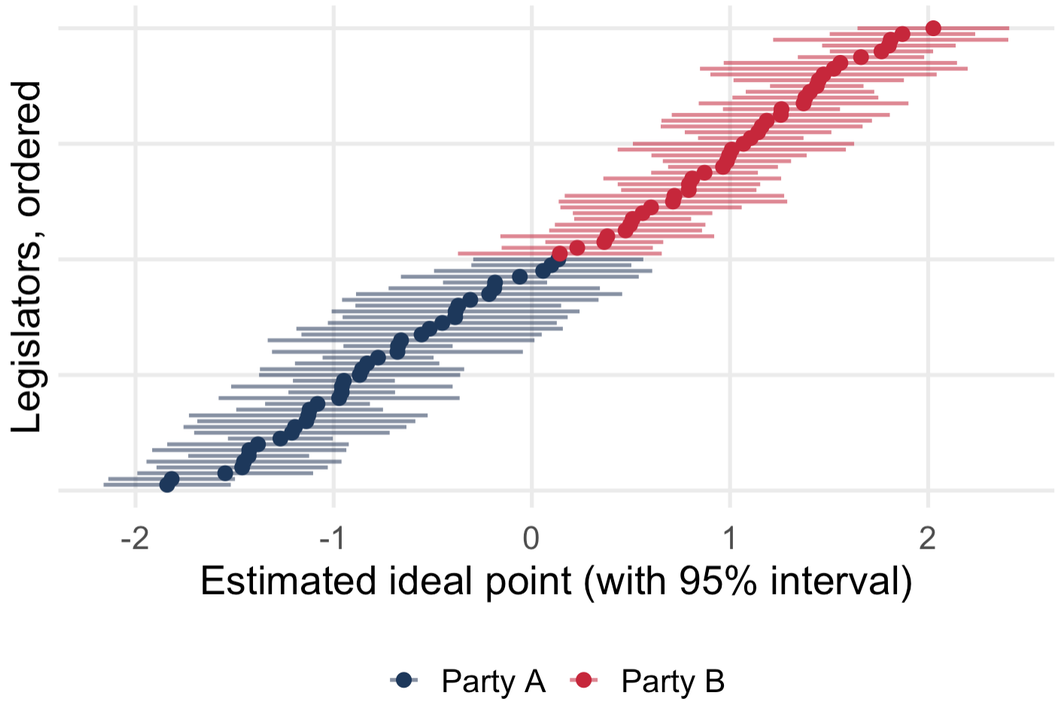
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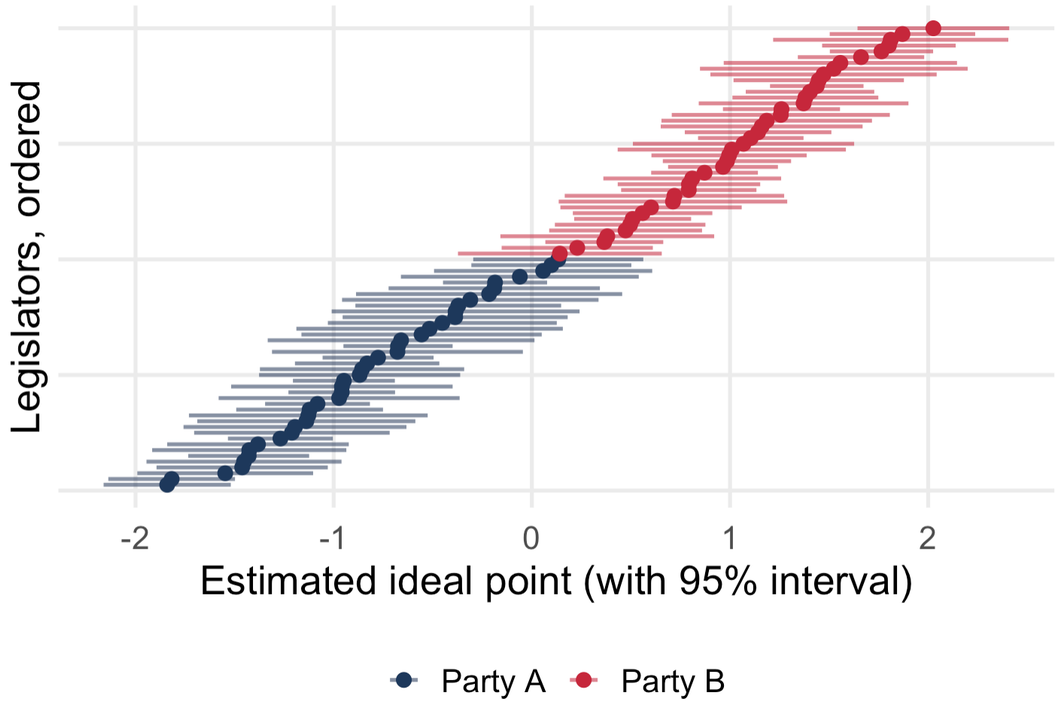
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- The estimated θ_i are **ideal points**. Calling the dimension “left–right” is a validity claim, not something the model tells you.



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Does the measure travel?

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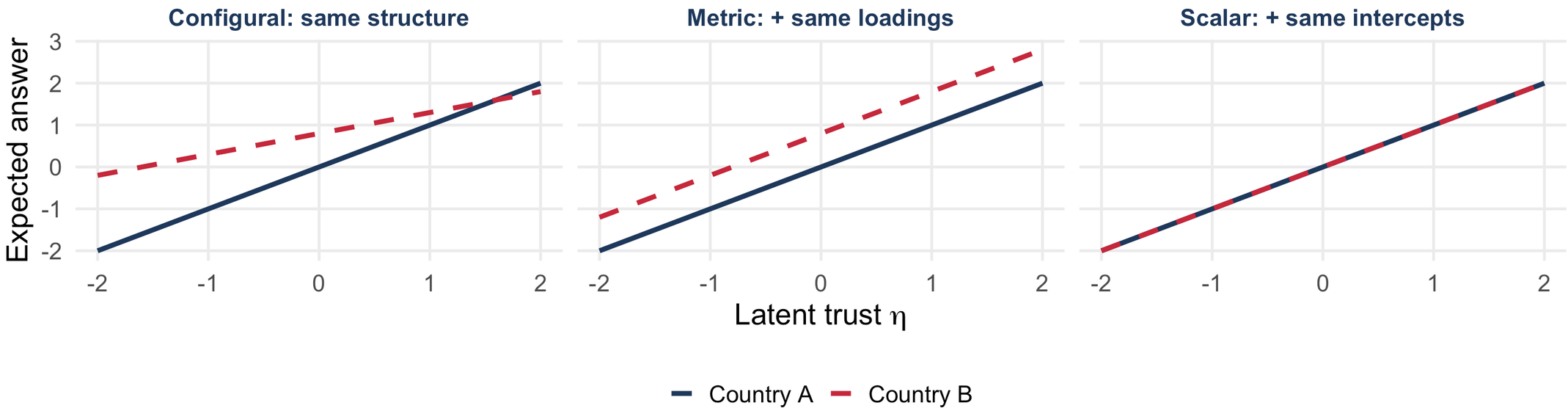
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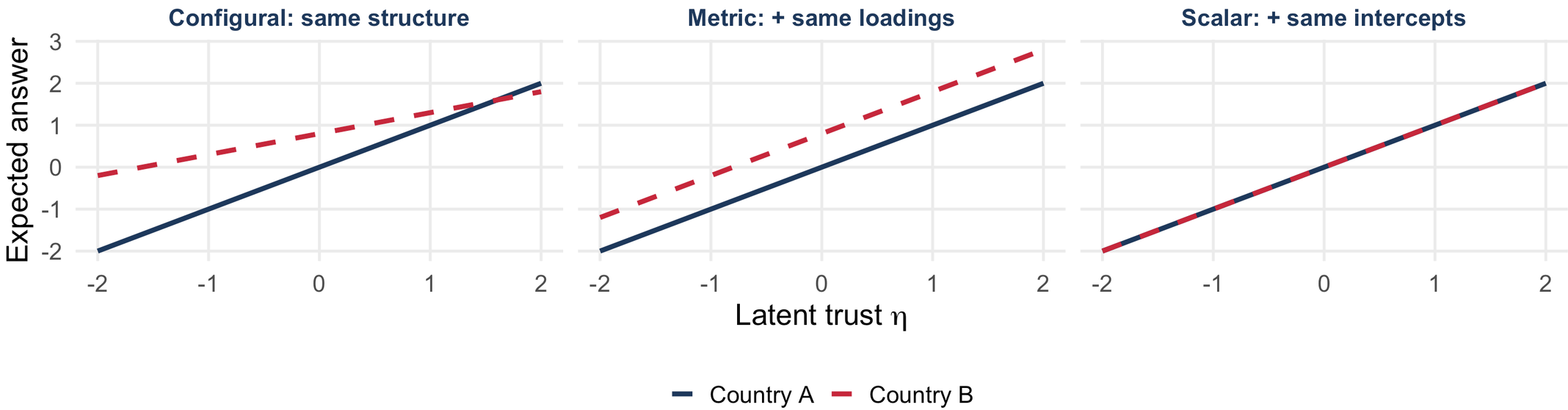
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None of these shows up as an error in the data. They look like **findings**. **Measurement invariance** asks: do two people with the same level of the trait give the same answer, whatever group they are in?

Three levels of invariance

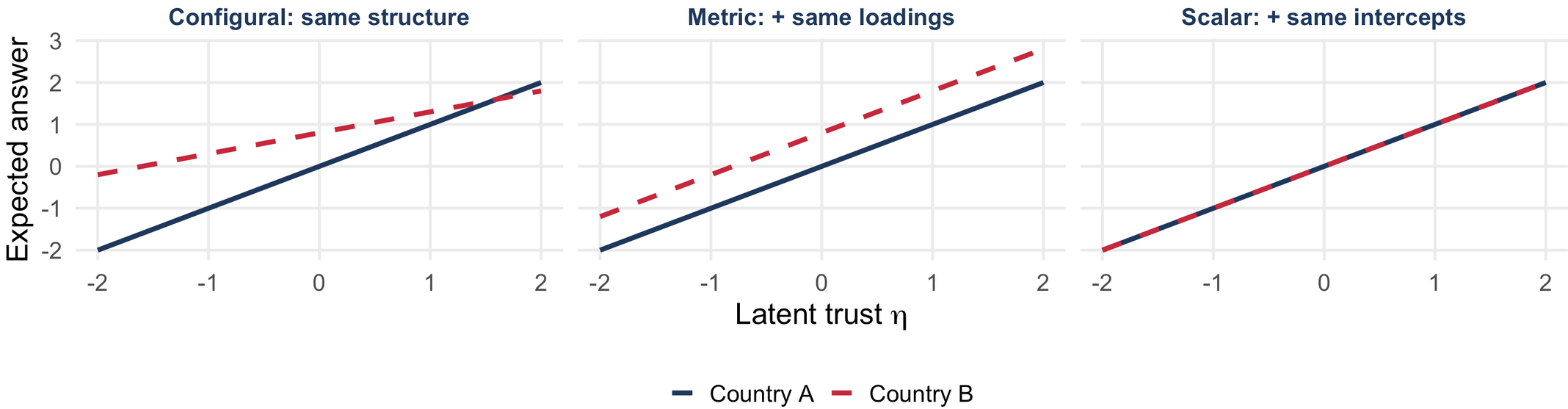


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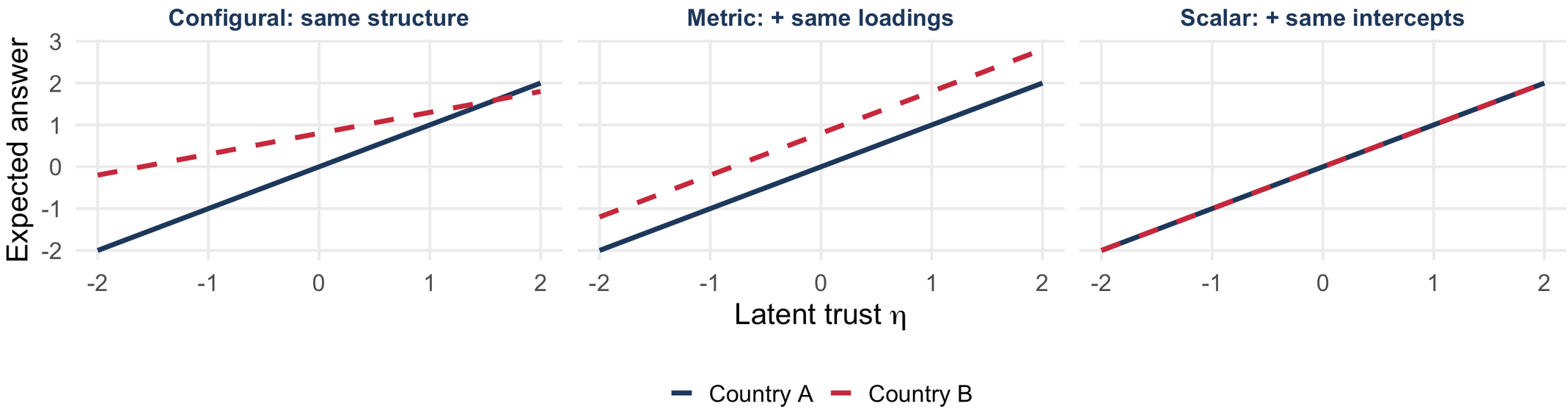
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Same unit **and** same zero: the same trust gives the same answer. Now you can compare **means**.

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A construct can travel while its measures, and its political meaning, do not.

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When scalar invariance fails in country number five.

Wrapping up

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- **Where is it precise, and is the uncertainty carried forward?** Information curves, not just one reliability number.
- **Does the measure mean the same thing in every group and year you compare?** At the level of invariance your comparison needs.

Your turn ✏️ (~10 min, in pairs, then regroup)

Sketch the bones of a **measurement paper** for your PhD. Start from last week's concept, or pick a new one.

1. **Construct**: what do you want to measure, and why do existing measures fall short? How many dimensions does theory say?
2. **Items**: write two or three. Do they reflect the construct, or form it?
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Regroup: two or three pairs pitch their paper in one minute. The room names the biggest threat to it.

What we covered today

FROM MANY INDICATORS TO ONE (OR TWO) CONSTRUCTS

Latent variables and factor analysis

A construct causes its indicators (local independence); PCA compresses, factor analysis measures; EFA explores, CFA tests competing theories on new data

Item response theory

Difficulty, discrimination, guessing; 1PL, 2PL, 3PL; information shows where a measure is precise, and purpose decides where it should be

Rules of thumb are choices

Cutoffs for loadings, fit, and discrimination are conventions: let theory lead, fix the rules in advance, report what moves

Does the measure travel?

Configural, metric, scalar invariance; a construct can travel from Wisconsin to Europe while its scale and its politics do not

Readings

Compulsory

- Campos & Federico (2026). A new measure of affective polarization. *APSR* 120(1): 160–178.
- Claassen et al. (2025). Measuring rural and urban consciousness in Europe. *Electoral Studies* 95: 102912.

Strongly encouraged

- Davidov, Meuleman, Cieciuch, Schmidt & Billiet (2014). Measurement equivalence in cross-national research. *Annual Review of Sociology* 40: 55–75.

Optional

- Claassen et al. (2025). Conceptualizing and measuring support for democracy. *Comparative Political Studies* 58(6): 1171–1198.
- Treier & Jackman (2008). Democracy as a latent variable. *AJPS* 52(1): 201–217.
- Jackman (2008). Measurement. *Oxford Handbook of Political Methodology*.
- DeVellis & Thorpe, *Scale Development*, Ch. 1–8.

On measuring trust

- Bauer & Freitag (2018). Measuring trust. In Uslaner (Ed.), *The Oxford Handbook of Social and Political Trust*. Oxford.
- Hooghe (2011). *British Journal of Politics and International Relations* 13(2): 269–275.
- Rothstein & Stolle (2008). *Comparative Politics* 40(4): 441–459.
- Delhey, Newton & Welzel (2011). *American Sociological Review* 76(5): 786–807.
- Glaeser, Laibson, Scheinkman & Soutter (2000). *Quarterly Journal of Economics* 115(3): 811–846.

Also cited today

- Blackwell, Honaker & King (2017). *Sociological Methods & Research* 46(3): 303–341.
- Clinton, Jackman & Rivers (2004). *APSR* 98(2): 355–370.
- Cramer (2016). *The Politics of Resentment: Rural Consciousness in Wisconsin and the Rise of Scott Walker*. University of Chicago Press.
- Fariss (2014). *APSR* 108(2): 297–318.
- McNeish (2018). *Psychological Methods* 23(3): 412–433.
- Meredith (1993). *Psychometrika* 58(4): 525–543.
- Poznyak, Meuleman, Abts & Bishop (2014). *Social Indicators Research* 118(2).
- Ramirez-Ruiz & Munzert. Measuring scientific evidence consumption literacy for public policy with the INSPIRE inventory. Working paper.
- Rönkkö & Cho (2022). *Organizational Research Methods* 25(1): 6–47.
- Samejima (1969). *Psychometrika Monograph Supplement* 17.
- Coenders, Lubbers & Scheepers (2005) and Billiet (2013), on the Danish “crime” item: as reviewed in Davidov et al. (2014), which has the full references.
- PCA illustrations: Allison Horst (@allison_horst), CC BY 4.0.

A latent score is a theory, an estimate, and a claim.

Test the theory, carry the uncertainty,

and check the claim travels.

Next week: Maximum likelihood · binary outcomes

In the lab on Monday

with Martín

In Lab 5 you'll return to the trust battery: run an EFA, compare one- and two-factor CFAs, and compute ω. Then you'll fit IRT models to a knowledge quiz and build short forms from their information curves. An optional part on measurement invariance is there for you to try on your own.

`psych::fa()` `lavaan::cfa()` `mirt::mirt()` `ggmirt::tracePlot()`

LAB 5

`05-latent-variables-lab.qmd`